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Production and operations management for intelligent manufacturing: a systematic literature review

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ABSTRACT
In the context of Industry 4.0, the manufacturing sector is moving from automation towards intelligence. The application of new generation information and communication technologies (ICTs) improves the interconnection and transparency of intelligent manufacturing (IM) systems, which will change how information interacts and work is done, thus changing how work should be managed. These changes require the following characteristics for IM production and operations management (POM): integration, flexibility and networking, autonomous and collaborative decision-making, learning-based operations management, self-optimisation and adaptability, and proactive decision-making. This paper presents the state of the art, current challenges, and future directions of IM-related POM research from the perspectives of these characteristics through a systematic literature review. Descriptive and thematic analyses of 208 research articles published between 2005 and 2020 are provided. The review and discussions focus on five research themes, i.e. value creation mechanisms, resource configuration and capacity planning, production planning, scheduling, and logistics.

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KEYWORDS
Intelligent manufacturing; production and operations management; value creation mechanisms; resource configuration and capacity planning; production planning; scheduling; logistics

1. Introduction

In the increasingly competitive globalised market, the manufacturing sector is facing unprecedented challenges, such as rapidly changing personalised demands (Yang et al. 2017), increased supply chain complexity and global fragmentation of production (Hadar and Bilberg 2012; Burke et al. 2017), growing competitive pressures from unexpected sources (Burke et al. 2017), and talent-related challenges (Elejalde-Ruiz 2016). With the rapid development of automation technology, robotics, and new generation information and communication technologies (ICTs) such as the Internet of Things (IoT), 5th generation (5G) mobile networks, big data, cloud computing, digital twins, and artificial intelligence (AI), the manufacturing sector is moving from automation towards intelligence to address these challenges (Wang, Wan, et al. 2016; Burke et al. 2017; Zhong et al. 2017; Egger and Masood 2020). The German government has proposed the Industry 4.0 strategy, which is to build a cyber-physical system (CPS) based on ICTs to seamlessly connect the physical and digital world and thus improve the adaptability, autonomy, flexibility, and intelligence of manufacturing systems (Egger and Masood 2020). The main economies around the world are vigorously promoting or revitalising the manufacturing sector and have promulgated development strategies related to intelligent manufacturing (IM), including the US Advanced Manufacturing Partnership, China's Made in China 2025, Japan's White Paper on Manufacturing Industries, the EU's Europe 2020, the UK's Future of Manufacturing, and Korea's Public-Private Joint Smart Factory Promotion Team.

IM is a broad concept of manufacturing that has been under continuous development for decades (Kusiak 1990; Rzevski 1997; Oztemel 2010; Hozdić 2015; Burke et al. 2017; Zhong et al. 2017; Zhou et al. 2018). Some paradigms, such as smart manufacturing, are frequently used synonymously with IM (Zhong et al. 2017; Mittal et al. 2019). In this paper, we follow the literature to use IM because more publications use this term (Wang, Tao, et al. 2021). IM is an agile manufacturing system enabled by ICTs that can enable vertical and horizontal integration, self-optimisation across a wider network, adapting and learning from new situations, and autonomous operations to continuously raise product quality, performance, and service levels while reducing cost and resource consumption (Hozdić 2015; Burke et al. 2017; Zhong et al. 2017; Zhou et al. 2018). IM systems such as smart factories are one way to realise IM (Zhong et al.

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2017). Although the perfect IM system that achieves all of the above goals may not have been realised, there are numerous IM applications in the manufacturing industry that achieve some of the goals. For example, GE invested over $200 million in a smart factory in Pune, India, which enhances flexibility to quickly adjust to changing demands; Faurecia developed a smart factory in Columbus, Indiana, which enables self-learning autonomous intelligent vehicles, collaborative robots, and continuous data collection; and Audi established a smart factory in Mexico (Rossmann et al. 2017).

The key features of IM are connectivity, optimisation, transparency, predictive, and agility (Daryll et al. 2015; Burke et al. 2017). These features help IM systems adapt to changes in the internal and external environment and support more rational management decisions. According to a study by Capgemini Consulting, IM could add between $500 billion and $1.5 trillion in value to the global economy in 5 years, accelerate the on-time delivery of finished products and greatly improve quality indicators (Rossmann et al. 2017).

To realise the above features and desired benefits of IM, it is critical to propose adapted production and operations management (POM) approaches based on the advantages brought by ICTs with the following key characteristics.

- **Integration:** Horizontal integration of supply chain systems within and across factories and vertical integration through connected manufacturing systems on physical facilities and human assets drive the digitisation of manufacturing, maintenance, and inventory tracking and operations across the entire network.
- **Flexibility and Networking:** Flexibility means that the production system is able to execute different production processes, requiring very high availability of processing units. Networking means that all information related to product manufacturing is shared in real time between different systems, especially information about orders, personnel, and resources, to achieve synergy in the manufacturing network.
- **Autonomous and Collaborative Decision-Making:** IM value chains are distributed and dependent on complex information and logistics. In a decentralised decision model, each independent entity makes its own decision, requiring advanced algorithms for individual local optimisation and global collaborative optimisation for faster and more efficient responses to unforeseen changes.
- **Learning-Based Operations Management:** IM machines are no longer just objects to be managed, but can also learn from completed tasks and optimise their settings. Machines respond to events at a higher level of cognition and gain intelligence over time, allowing production control to become self-running and agile.
- **Self-Optimisation and Adaptability:** Thanks to powerful computational and analytical capabilities, the IM control system can gain real-time flexibility to adapt to production without human interaction or interruption of the process. It has the ability to self-optimise to adapt to changes that were previously difficult.
- **Proactive Decision-Making:** It is costly and time-consuming to respond passively to changes and system disturbances. IM systems need the ability to anticipate and take action to prevent problems, including identifying exceptions, replenishing inventory, predicting quality issues, and monitoring safety and maintenance issues.

In the past two decades, IM has emerged as an important research area and an increasing number of scholars and companies are creating related research and applications. Many systematic reviews in this field have been carried out in recent years, and they can be divided into the following three categories.

(1) Related technologies to support IM

This literature stream reviews the studies on ICTs in support of IM. Tian, Yin, and Taylor (2002) review advanced information technology and system and manufacturing automation and outline the infrastructure of internet-based manufacturing. Mittal et al. (2019) identify the characteristics, technologies, and enabling factors related to IM. Yang et al. (2019) present a review of IoT technologies and systems that are the drivers and foundations of IM. Egger and Masood (2020) and He and Bai (2021) review the research on augmented reality and digital twins for supporting IM, respectively. Olsen and Tomlin (2020) describe the technologies inherent in Industry 4.0 and discuss the opportunities and challenges for research in this area, including additive manufacturing, IoT, blockchain, advanced robotics, and AI.

(2) Focusing on a specific area of IM

Some papers focus on a specific area of IM. Shen et al. (2006) provide a review on the agent-based systems in IM. Li, Hou, et al. (2017) are more concerned with AI technology in IM. Meng et al. (2018) review the sustainability and energy efficiency in IM systems. Besharati-Foumani, Lohtander, and Varis (2019) focus on the review of intelligent process planning approaches for IM systems. Ren et al. (2019) conduct an overview of big data analytics in IM. Kamble et al. (2020) perform a literature review to identify performance measures of IM systems and develop a performance measurement system framework.

(3) General reviews of IM

A large number of papers generally review the literature on IM in the context of Industry 4.0 (Kang et al. 2016; Jardim-Goncalves, Romero, and Grilo 2017; Zhong et al. 2017; Kusiak 2018; Zheng et al. 2018; Osterrieder, Budde, and Friedli 2020; Wang, Tao, et al. 2021). In these papers, IM systems and problems are discussed in terms of definitions and concepts, structures and frameworks, classifications and future trends, theories and models, scenarios and applications, challenges, innovations, related techniques, and strategies. Such papers are useful to provide a broad understanding of IM.

In the current stage with relatively mature ICTs, one of the pressing needs of manufacturers is how to make intelligent POM decisions, including the mechanism, evaluation, configuration, organisation, planning, and scheduling decisions, for the effective and efficient management of IM systems (Zhong et al. 2017; Olsen and Tomlin 2020). However, to our knowledge, there is no review paper that adequately discusses POM studies in IM and considerable effort is still needed to improve operations management of IM systems. The main objectives of this paper are to review and summarise the existing related literature on POM studies in IM to identify the recent trends, analyse the limitations and challenges of implementing key characteristics of IM systems, and propose future directions. Studies using POM approaches to achieve the key characteristics of IM are the focus of this paper.

The main tasks for POM in the manufacturing system are the activities involved in transforming a product idea into a finished product, as well as those involved in planning and controlling the systems that produce goods, which led to a wide range of research topics and themes such as value creation, resource organisation, production planning and control, scheduling, logistics (Kouvelis, Chambers, and Yu 2005; Gupta, Verma, and Victorino 2006; Merigó et al. 2019; Guo, Li, et al. 2020). Thus, this paper reviews the related studies in terms of five themes: the value creation mechanism, resource configuration and capacity planning, production planning, scheduling, and logistics. Our proposed review framework for considering the POM literature for IM is shown in Figure 1. It also shows the interactions between those themes and their main contents. The value creation mechanism is the intrinsic motivation for activities in the IM system and thus the basis for other research themes. Studies in this area form our first focus based on a framework of the factors that drive the implementation of IM and affect its application and outcomes. We then analyse the literature on resource configuration and capacity planning, production planning, scheduling, and logistics from different perspectives, including the framework, methodology, and the scopes and features. Finally, we summarise the current state of the literature that considers these five characteristics, discuss the opportunities and challenges, and identify potential future directions for POM research in this area.

The rest of the paper is organised as follows. Section 2 describes the methodology applied in this review paper. Section 3 details the descriptive analyses, and Section 4 presents the thematic analyses of the POM studies based on the five themes mentioned above. Section 5 discusses the current state of research as well as limitations and potential future directions of the field. Section 6 presents the concluding remarks.

2. Methodology

This paper aims to synthesise the existing literature and identify directions for future research in IM POM studies. A systematic review was carried out to provide a holistic, state-of-the-art view of research development in this area. Following the generic framework for a systematic review by Tranfield, Denyer, and Smart (2003), we adopted a three-step approach: (a) planning, (b) conducting, and (c) reporting and dissemination. The planning of the review was presented by introducing the framework in the previous section. This section presents how the review was conducted, including the data identification, data extraction, and data synthesis. The reporting and dissemination of the data are presented in Sections 3 and 4.

To retrieve as many relevant papers as possible, we performed searches in six databases: Web of Science (WoS), ProQuest, ScienceDirect, EBSCO, Wiley, and IEEE Xplore. These databases include most of the publishers and high-quality journals in this area. IM is a popular topic, and a search for this keyword yielded more than 40,000 results in WoS; we therefore combined the keyword IM or its synonyms with some keywords related to the five POM research themes to focus on the scope of this review paper. The keywords used for the search in each theme are listed in Table 1.

To maximise knowledge discovery and theme convergence, we screened the 'grey' literature and focused on articles from quality journals published in English. The literature related to IM has been growing steadily since 2005 (Zhong et al. 2017), so our search period ranged from 2005 to 2020. Figure 2 shows the number of results from each database. For all databases, the keywords were applied to the title, abstract, and keywords. A

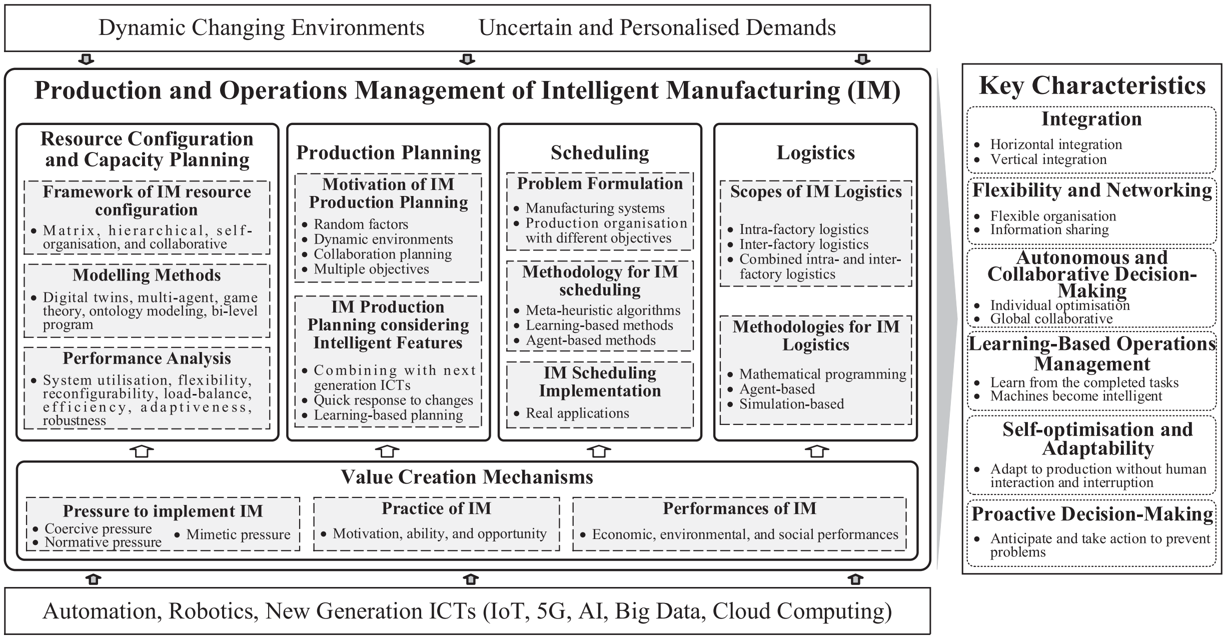


Figure 1. Review framework of POM in IM.

Table 1. Keywords used in literature search for IM POM by theme

Themes	Keywords
Value creation mechanisms	('IM' or 'smart manufacturing' OR 'smart factory' OR 'intelligent factory') AND ('value' OR 'performance' OR 'outcome' OR 'effect' OR 'impact' OR 'benefit' OR 'consequence')
Resource configuration and capacity planning	('IM' or 'smart manufacturing' OR 'smart factory' OR 'intelligent factory') AND ('configuration' OR 'reconfiguration' OR 'capacity planning')
Production planning	('IM' or 'smart manufacturing' OR 'smart factory' OR 'intelligent factory') AND ('production planning' OR 'production plan')
Scheduling	('IM' or 'smart manufacturing' OR 'smart factory' OR 'intelligent factory') AND ('scheduling')
Logistics	('smart manufacturing' OR 'IM' OR 'smart factory' OR 'intelligent factory') AND ('logistics')

total of 2271 papers were collected by searching the above keywords in the six databases. After the elimination of duplicate reference papers from all of the databases, 2176 papers remained. We then read the papers and screened them by checking if they fit in the scope of our review based on the following criteria: in English; manufacturing and POM-related (mechanism, evaluation, configuration, organisation, planning, scheduling and control for the management of production and operations); and present a method rather than a review. After this procedure, 208 articles were ultimately selected (as shown in the Appendix).

3. Descriptive analyses

For the third step of our systematic review (i.e. the reporting and dissemination), the descriptive and thematic analyses are presented in this section and the next section, respectively. The descriptive analysis was performed based on the published year and research theme, authors' country, and industrial sectors.

3.1. Distribution by year of publication and research theme

This subsection analyses the identified papers based on the year of publication and the research theme to illustrate the trend of relevant research themes. Figure 3 shows an increasing trend of academic interest in IM POM, especially in the last three years. It is also evident that the research on the five themes follows a similar trend. The growing trends correspond to the increasing maturity and application of ICTs and the increasing emphasis on IM strategies in various countries. From the perspective of the research themes, the two themes of value creation mechanisms and scheduling have received more attention than the other three themes. Note that in Figure 3, the total number of papers on the five research themes (209) is larger than the number of selected papers (208) because some papers consider more than one theme.

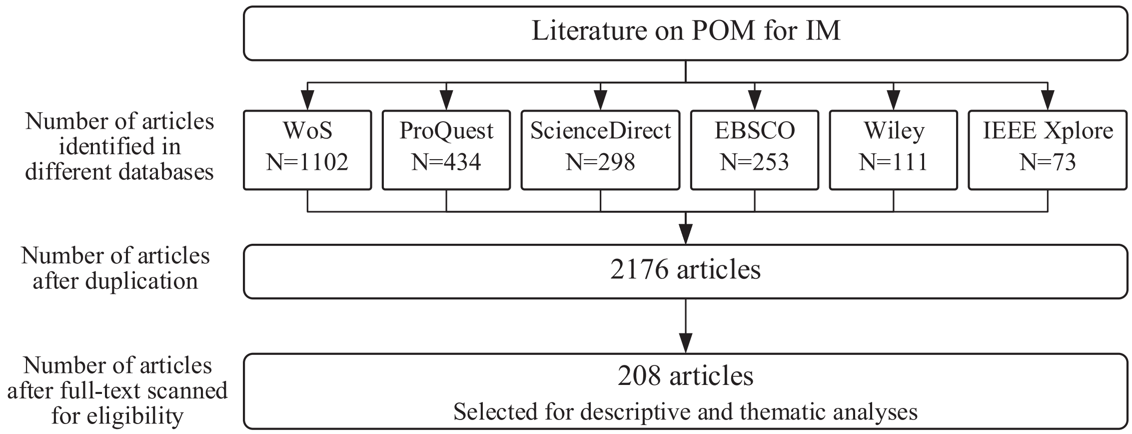
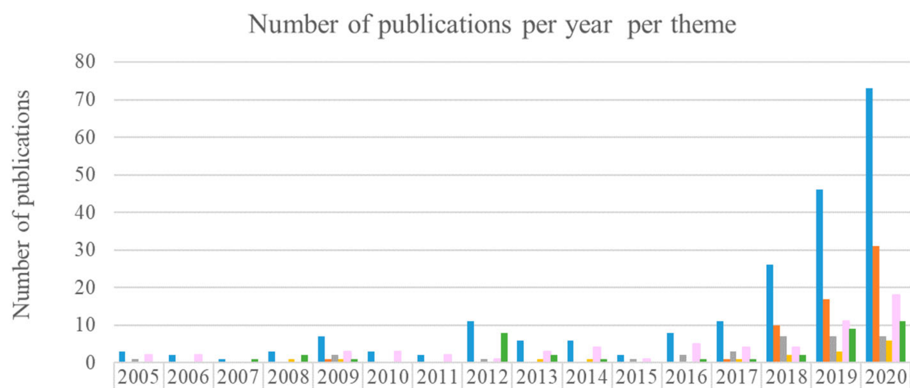


Figure 2. Data collection for the systematic literature review.



	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Total number	3	2	1	3	7	3	2	11	6	6	2	8	11	26	46	73
Value creation mechanisms	0	0	0	0	1	0	0	0	0	0	0	0	1	10	17	31
Resource configuration and capacity planning	1	0	0	0	2	0	0	1	0	0	1	2	3	7	7	7
Production planning	0	0	0	1	1	0	0	0	1	1	0	0	1	2	3	6
Scheduling	2	2	0	0	3	3	2	1	3	4	1	5	4	4	11	18
Logistics	0	0	1	2	1	0	0	8	2	1	0	1	1	2	9	11

Figure 3. Paper distribution by year of publication and research theme.

3.2. Distribution by country

This subsection investigates the distribution of authors' countries of origin to clarify the current state of IM POM research in different regions and countries. Figure 4 illustrates the distribution of countries of the first affiliation to which the first author of the papers belongs. It shows two dominant categories: Asia and Europe, and more than 86% of papers have their origins in Asia and Europe. This reflects the prominence of these two regions in IM POM research. This prominence can be ascribed to the fact that the IM-related concept Industry 4.0 first originated in Europe, so European scholars and manufacturers have made pioneering efforts in promoting IM. Elsewhere, Asia has many scholars conducting related research, perhaps because it has many manufacturing countries such as China, Japan, and South Korea, and these countries have proposed IM-related strategies, thus attracting many scholars to conduct research in this area; for example, China has contributed to more than half of the papers from Asian countries.

3.3. Distribution across manufacturing industrial sectors

To determine the distribution of practical applications, we classified the manufacturing industries primarily involved in the identified papers. The classification criteria for manufacturing industries are based on the 2012 North American Industry Classification System by Statistics Canada. Papers that mention more than one industry

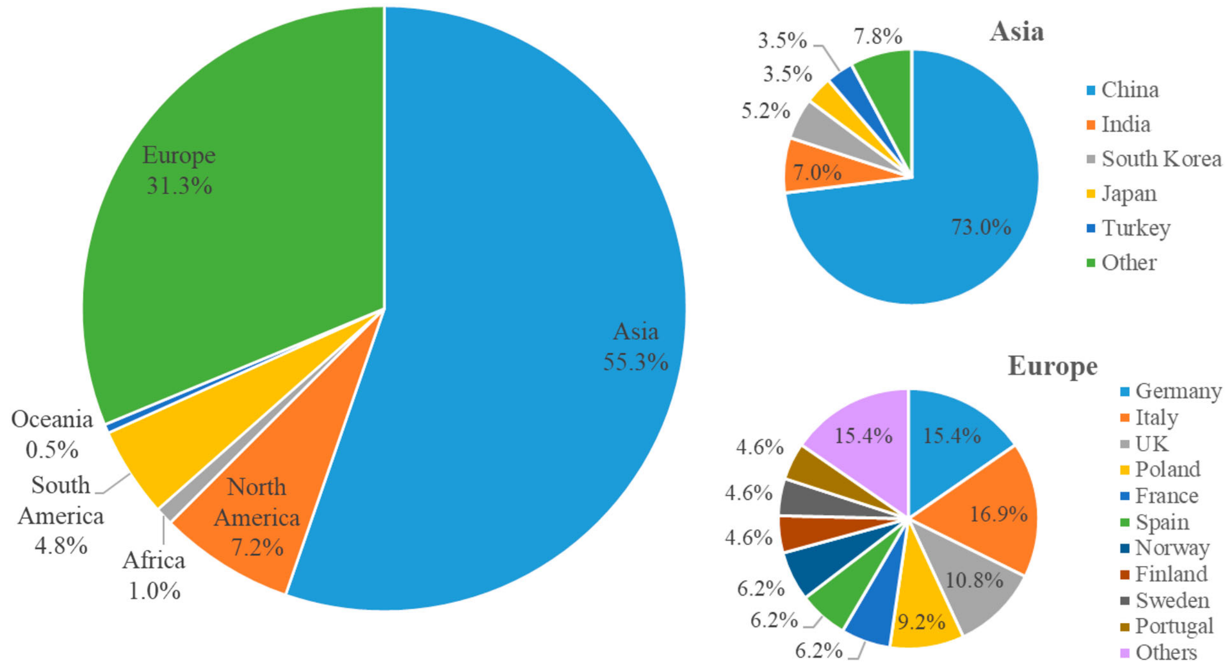


Figure 4. Paper distribution by country.

are classified as diverse, whereas those that do not mention a specific industry are classified as general. Figure 5 shows that more than half of the papers do not mention a specific industry. This may be because many of the papers do not focus on a practical case or industry but are general studies considering IM-related features and scenarios. For the manufacturing industrial sectors, electrical equipment, computer and electronic products, transportation equipment, and machinery manufacturing are the most studied. This can be attributed to their relatively high degree of automation, facilitating the transition to IM. In addition, the percentage of scheduling-related studies that fall in the category of general is 87.3%, which indicates that fewer papers on this theme specify a specific industry. This might be because the relevant methods for scheduling are relatively generic across industries and do not need to emphasise specific industries.

4. Thematic analyses

This section provides a thematic analysis of the 208 selected papers in terms of the five POM research themes presented in the review framework.

4.1. IM value creation mechanisms

Several publications have indicated that IM, instead of a single transformation of the production mode, is an implementation of collective embedded technologies and knowledge to pursue the integration of interconnected data and optimisation of value chains (Martínez-Olvera and Mora-Vargas 2019; Bokrantz et al. 2020b). However, no single research has integrated several scopes to explain exactly how IM creates value for manufacturers. Therefore, we approach the issue through an analysis of value creation mechanisms of IM. Since the value creation process of IM are not thoroughly understood by academia, we require an instrument to direct us to reveal the value creation mechanisms for the IM. The pressure-practice-performance (PPP) framework fits our intention and provides a suitable conceptual framework for us to compare the adoption and influence of the IM (Zhu and Sarkis 2007; Zhu, Crotty, and Sarkis 2008). Specifically, the PPP framework is widely used in literature reviews and empirical studies to reveal the drivers, applications, and possible outcomes of an emerging phenomenon. The three aspects depict the developing stages of a new phenomenon, revealing why a phenomenon takes place, how it affects the management applications, and what outcomes it brings to relevant firms or industries. For instance, Zhu, Sarkis, and Geng (2005) and Zhu, Sarkis, and Lai (2007) use this framework to illustrate how green supply chain management developed in China by indicating the organisational pressures, practice behaviours, and economic benefits of emergent green supply chain management. Zhu, Crotty, and Sarkis (2008) characterise the pressure, adoption practice and performance

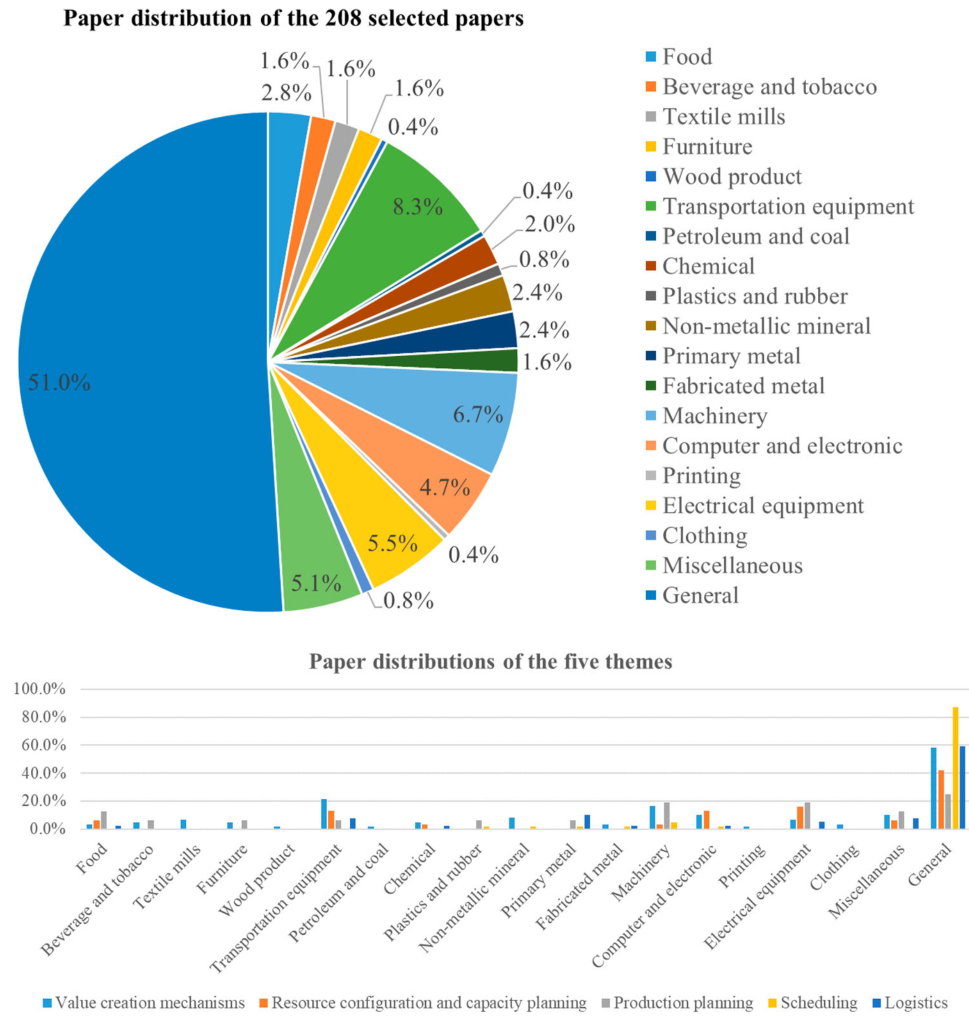


Figure 5. Paper distributions across industrial sectors.

of environmental supply chain management in different countries. Lu et al. (2018) model institutional pressures, firm practices, and performance outcomes for the sustainable supply chain management. In this research, we establish a PPP framework to characterise the factors that drive IM implementation, affect its application process, and illustrate its possible outcomes. Fragmented findings are consolidated according to the three categories, as shown in Figure 6, and the relative subcategories further clarify the different PPP levels.

4.1.1. Pressure to implement IM

As for the pressure to implement IM, we apply institutional theory to illustrate how the three external institutional isomorphisms, i.e. coercive, normative, and mimetic pressures, force manufacturers to implement IM (DiMaggio and Powell 1983; Jabbour et al. 2019; Fuchs 2020; Bokrantz et al. 2020b).

Coercive pressure refers to certain institutions' regulatory enforcement on the adoption of IM. For economic and competitive purposes, governments around the world have realised the importance of IM. They either propose initiatives and action plans to promote nationwide IM-oriented transformation, or generate specific investment, tax benefits, policies, or sponsorships, for specific manufacturers and industries to develop IM (Büchi, Cugno, and Castagnoli 2020). German Industry 4.0 strategy initialise the regulatory force on the digitisation of production processes, smart manufacturing, and inter-company connectivity (Mueller, Buliga, and Voigt 2018). German recent High-Tech Strategy 2020 provides specific action plans for manufactures' digitised and intelligent transformation in the production and operations process (Al-Sayed and Yang 2020; Buer et al. 2021). Chinese government has also highly publicised the transformative application of intelligent technology and has launched ambitious strategies such as Made in China

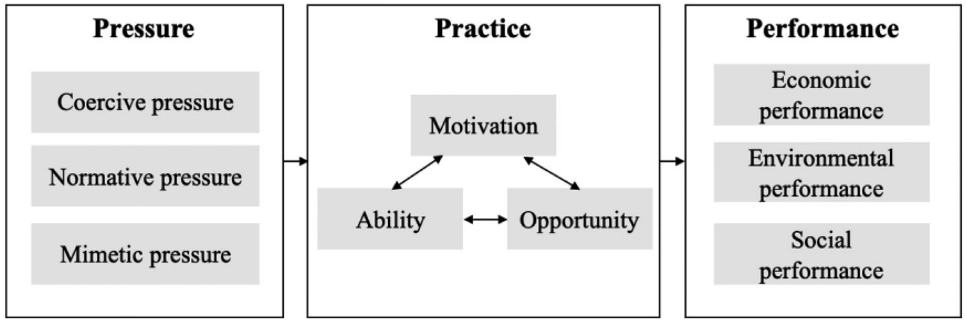


Figure 6. PPP framework for IM value creation mechanisms.

2025 to accelerate the intelligence-oriented transformation of manufacturing processes (Al-Sayed and Yang 2020). Advanced Manufacturing Partnership in the US, Future of Manufacturing in the UK, The Declaration on Creating the Best IT Country in Japan, and Public-Private Joint Smart Factory Promotion Team in Korea provide similar pressures force manufacturers to construct intelligent technology infrastructure, ecosystems, and industrial clusters (Won and Park 2020; Zhang, Ming, and Yin 2020; Bokrantz et al. 2020a).

Normative pressure refers to manufacturers' adoption of IM to fit with prevailing norms, values, and social expectations (Saniuk, Grabowska, and Gajdzik 2020). Normative pressure comes from the implicit or explicit lobbying efforts of professionals and other actors (Meyer and Rowan 1977). Publications on an upcoming IM revolution from an influential research institute, technology provider, or consulting company (e.g. McKinsey Co.) increase the manufacturers' awareness, the perceived benefits, and the social expectation on such new manufacturing mode (Cordeiro, Ordóñez, and Ferro 2019; Mittal et al. 2020; Won and Park 2020). Their broadcast on the key structure and priorities of IM elements can be used as the IM benchmark for practitioners (Jerman, Bach, and Aleksic 2020). The normative pressure of these organisations is so profound that even 14 papers in our pool use either the definition in McKinsey's white book or its statistic results as one of the basic materials that define the state-of-art of IM (Sanghavi, Parikh, and Raj 2019; Jerman, Bach, and Aleksic 2020; Saniuk, Grabowska, and Gajdzik 2020). These publications further broadcast the popular idea of smart technology and IM, and thereby motivate the forward-looking intentions of the consumer, producer, and employee to expect the industry to take actions to change by implementing IM (Bokrantz et al. 2020a).

Pressures to mimic other companies to keep up with the market trends are also prevalent. Uncertainty and anxiety on the response to the industrial trends may force manufactures to mimic or replicate the IM strategy of competitors or companies from the same industry (Trstenjak et al. 2020). Industry leaders' implementation of IM, for example, Microsoft's adoption of augmented reality camera in the production (Brizzi et al. 2018; Mueller, Buliga, and Voigt 2018), generates an implicit signal for the advantage of such technology and drives the whole industry to follow the IM vision and transformation strategy (Jones et al. 2020; Kovacova et al. 2020; Trstenjak et al. 2020). Bokrantz et al. (2020a) further divide the IM mimicry into three modes: frequency-based (mimic actions taken by many companies), trait-based (mimic companies with outstanding traits such as large firm size), and outcome-based (mimic actions that have proved successful).

Hence, three institutional pressure coexists during the adoption of IM. Manufacturers implement IM to achieve legitimacy, obtain the first-mover advantages, or maintain the industry position. Among the three, some publications believe coercive pressures often appear along with normative ones for an individual practitioner (Bokrantz et al. 2020a).

4.1.2. Practice of IM

As manufacturers around the globe gradually implement the IM strategy in different norms and paths, these actions need to be carefully examined to give insights into how such strategy could benefit the manufacturers in practice. We refer to the motivation-ability-opportunity (MAO) framework, which is a theoretically-rich framework used for analysing organisational and consumer actions and their outcomes in marketing and other management areas (Batra and Ray 1986; MacInnis, Moorman, and Jaworski 1991; Hallahan 2000). Studies in recent years also use this framework to analyse the practices and effectiveness of supply chain strategy or organisational transformations and to offer management mechanisms for the industry (Argote, McEvily, and Reagans 2003; Reinholt, Pedersen, and Foss 2011; Kim, Hur, and

Schoenherr 2015). We view the MAO framework as a theoretical perspective that guides us to investigate how manufacturers take actions during the IM implementation to generate value in three aspects (Hallahan 2000): within a focal firm, which triggers manufacturers to invest in the IM strategy (motivation), how manufacturers allocate their capability in developing IM-related transformation in reality (ability), and which conditions or aspects in the IM strategy have not been widely explored yet (opportunity). Specifically, we would provide the factors that strengthen a manufacturer's determination in IM adoptions, the activities and resources that are necessary for IM implementation, and the environmental conditions that provide perfect basis prerequisites for IM transformation, based on the literature.

Motivations to implement IM. Compared to the three external forces in section 4.1.1, we then list what internal factors and conditions have motivated an individual company to spontaneously start the IM transformation. First, IM is motivated by a high level of business relevance. A manufacturer may create an IM tailored vision when the existing products are required to shift from data generation to data transmission, the businesses are about to be expanded to cover the market niche, or the company plans to link up to global value chain (Fuchs 2020; Mittal et al. 2020). IM adoption essentially assists the business transformation, especially for the small and medium enterprises (Mittal et al. 2020). Second, since the implementing IM alone is costly, manufacturers are motivated to implement IM when it is inconsistent with other firm strategies and IM help unleash the potential of other strategies, such as mass customisation, lean manufacturing, circular economy, and short product life cycles, that are more directly relevant to revenue output (Jabbour et al. 2019; Buer et al. 2021; Fuchs 2020; Shi et al. 2020). Third, a manufacturer initialises the IM transformation only when the strategy is under controllable risks. Publications found that manufacturers conduct readiness factor calculations to recognise the current state of digitalisation extent, develop adequate IM investment plans, and avoid being distorted in virtual environments (Brizzi et al. 2018; Trstenjak et al. 2020). Manufacturers would ensure the system performance (e.g. accuracy, efficiency, and reliability) and the easily infusion with digital technologies to realise modularity, interoperability, virtualisation, real-time control, and service orientation, etc. (Brizzi et al. 2018; Seetharaman et al. 2019; Mittal et al. 2020; Shi et al. 2020).

Abilities to implement IM. We delineate the enablers (abilities) for the value realisation of IM. In general, IM is a value aggregation system consisting of a complex combination of operationalisation resources, business models, organisational layout, manufacturing knowledge, and management capability (Cordeiro, Ordóñez, and Ferro 2019; Shi et al. 2020; Wagire, Rathore, and Jain 2020). Since several literature reviews have emphasised the enabling technologies for IM (Culot et al. 2020; Wagire, Rathore, and Jain 2020), we aggregately summarise the key IM-enabling tangible and intangible business abilities from the firm level to the supply chain level in Figure 7.

Four clusters of IM enablers are identified. The left part of the business abilities in Figure 7 indicates the physical enabling hardware and software that construct the intelligent, connected, and automated infrastructure (Lin, Sheng, and Jeng Wang 2020). Since physical techniques and assets alone cannot realise the IM implementation (Qu et al. 2019), the right part illustrates the internal and external competencies that either assist or complement the physical assets to promote the repaid response, accurate reconfiguration, and smooth operations of the IM system.

(a) *Organisational physical assets* build up the fundamental structure for internal IM systems (Trstenjak et al. 2020). The smart machines and products drive the IM system to be intelligent-oriented, and thereby advanced robotics (Büchi, Cugno, and Castagnoli 2020), embedded AI wearables (Benotsmane, Kovács, and Dudás 2019; Yang, Ying, and Gao 2020), and process visualisation systems (Kamble et al. 2020) can interdependently supervise and control the production, quality certification, and logistics process so that autonomous decisions such as predictive maintenance and proactive planning can be accurately made in the workshop (Imran, ul Hameed, and ul Haque 2018; Qu et al. 2019; Lukasik and Stachowiak 2020). End-to-end integration ports unify the data transmission among different devices and enable horizontal integration between departments to either reduce time and cost or enhance intelligence and efficiency (Lafferty 2019; Sanghavi, Parikh, and Raj 2019).
(b) *Networking physical assets* realise the external vertical integration by connecting cross-enterprise devices with Ethernet, Wi-Fi, interfaces, high-frequency signals, Bluetooth, and cloud systems (Chu et al. 2020). IoT has been the main physical enabler of the constantly-feedback production and delivery process (Fatorachian and Kazemi 2018; Kao, Nawata, and Huang 2019b; Nimbalkar et al. 2020; Liu, Feng, et al. 2020), whereas CPS enables the communications between products, human, and machines by bridging cyberspaces and the physical worlds. Cloud-computing servers have created

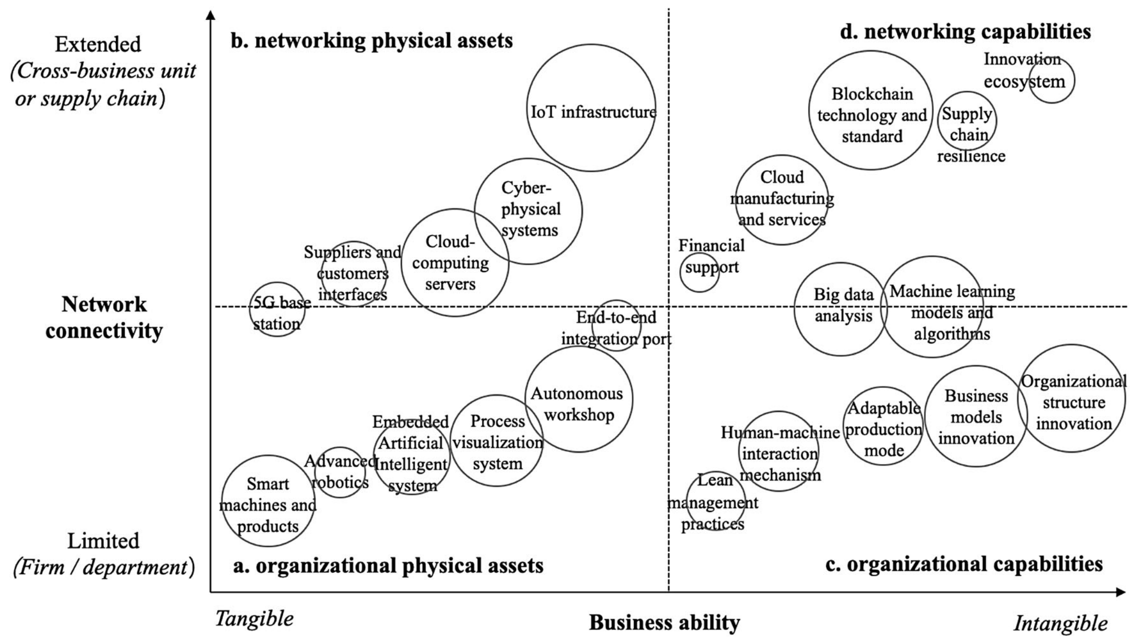


Figure 7. Key enabling business ability clustered by the nature of ability and level of network connectivity.

platforms specialised in analysing dynamic data at a high speed so that all information can be independently assessed at different locations but integrated to make decisions (Qu et al. 2019). Supplier and customer interfaces and 5G bearer networks provide application requirements for cross-business unit end-to-end integration (Chu et al. 2020).

(c) *Organisational capabilities* include the internal smart-oriented management techniques (Lin, Sheng, and Jeng Wang 2020). Organisational structure changes such as job breath and control, centralisation of decision-making, and the number of hierarchical levels (Cagliano et al. 2019; García-Muiña et al. 2020), and business models such as how to combine innovations, technology, and new selling approaches to facilitate the system applications and how to construct value proposition for the customers should be decided alongside the technologies selection (Koilo 2019; Kohtamäki et al. 2020). In the face of diverse rapidly-changing demands, an adaptable production that emphasises modularised, decentralised, service-oriented, and self-learning production is preferred (Jerman, Bach, and Aleksic 2020; Won and Park 2020; Zhang, Ming, and Yin 2020). Constructing a human-machine interaction mechanism in the production process avoids simultaneous fatigue and promotes learning effects between the two (Digiesi et al. 2009). Additionally, several studies found that when lean production practices are used together with IM implementation, they have a synergistic effect that is greater than their individual effects (Buer et al. 2021; Ciano et al. 2021).

(d) *Networking capabilities* emphasise the competence and resources that are required to manage external complex environmental changes and networking relationships. Blockchain technology enabling dynamic and asymmetric information sharing among multiple stakeholders have become popular (Ge et al. 2020; Wang, Liu, et al. 2020). Cloud manufacturing is another way to transform the manufacturing capabilities encapsulated in the remote cloud to improve production efficiency (Wagire, Rathore, and Jain 2020; Liu, Feng, et al. 2020). Besides, since a manufacturer itself may not maintain competitive advantages in any global IM competition, searching for sufficient financial support for the IM transformation (Benotsmane, Kovács, and Dudás 2019; Al-Sayed and Yang 2020) and constructing a smart innovation ecosystem with academic intuitions, agencies, and other intermediates may nurture a long-term innovation, digestion, absorption, and re-innovation of IM (Kao, Nawata, and Huang 2019b; Al-Sayed and Yang 2020; Yang, Ying, and Gao 2020). However, although IM is attractive, public emergent events such as coronavirus pandemic emphasises the importance of supply chain resilience (Qu et al. 2019; Ramirez-Pena et al. 2020). IM practitioners should make a balance between exploration and exploitation.

Although these business assets are reported separately, they are deeply interdependent in practice (Culot et al. 2020). Physical assets are required to be matched with the suitable capabilities to unleash both tangible and intangibles abilities (Nimbalkar et al. 2020). Additionally, not all abilities are needed in any circumstance. Manufacturers should optimise the configurations between technological complexity and organisation transformation stages of the technology life cycle to avoid technology substitution (Moeuf et al. 2018; Cagliano et al. 2019).

Opportunities to develop IM. Even though the motivations and abilities are fundamental for IM, manufactures still wait for the appropriate environmental opportunities to release IM value creation: possible contingencies that affect IM value creation are revealed. Rapidly changing markets and the fast development of information technologies have brought opportunities for manufacturers to innovate new production modes, smart factories, and innovative mechanisms for IM (Shi et al. 2020). Therefore, we divide the possible opportunities into market opportunities and technology opportunities. We list the frequently-cited events and prevailing trends in Table 2 and summarise the catalytic factors that these events and trends generate for IM.

4.1.3. Performances of IM

This subsection analyses the possible outcomes based on the balance of the Triple Bottom Line and gain insight into whether publications have found that IM affects the economic, environmental, and social performance of manufacturers (Braccini and Margherita 2019). We respectively list the indicators for the three performances in Table 3, including both positive and negative outcomes.

In general, IM supports the Triple Bottom Line through the improvement of profitability, energy use, human rights, etc. This conclusion conforms to Braccini and Margherita (2019). However, most practitioners pay most attention to the economic value of IM implementation so that they may overlook the risks such as environmental degradation and short-term financial crisis (Braccini and Margherita 2019; Yang, Ying, and Gao 2020). Additionally, as the last subsection stated, due to the prevailing trends of green manufacturing and humanitarian relief, the environmental and social value of IM should be further investigated and qualified for managers to promote sustainable IM operations.

4.2. IM resource configuration and capacity planning

This subsection focuses on the literature on resource configuration and capacity planning in IM, which has been widely studied in the POM area. Traditional methods using centralised planning and control structures are too rigid and difficult to address the highly dynamic environment of IM. Distributed systems have been proposed to leverage their self-organising and flexible traits to manage highly volatile and complex scenarios (Ma et al. 2020). Figure 8 presents the review framework for this theme, which analyses articles from three perspectives: architecture, modelling methods, and performance.

4.2.1. Architecture of IM resource configuration

Many articles investigate the architecture of IM systems to ensure easy application and improve the efficiency of resource configuration. Schoenemann et al. (2015) propose the concept of **matrix-structured** manufacturing systems (MMS) to provide high operational flexibility and scalability by supplying redundant workstations for the same operations as well as a flexible product routing. They explain the main principles, elements, and control strategies of MMS, as well as presenting a simulation approach to evaluate MMS configurations. This architecture requires a large number of workstations and high costs, making it unsuitable for expensive smart devices.

Some papers develop **hierarchical architectures** for agent-based IM systems. Oztemel and Tekez (2009) and Tang et al. (2018) propose a reference model for integrated IM systems and cloud-assisted self-organised architecture to help smart agents to communicate and negotiate through networks, respectively. The agent interaction behaviour is modelled to structure the agents hierarchically to reduce the complexity for solving the challenge that the interactions among agents in distributed systems are difficult to understand and predict. This architecture can be easily deployed to build the IM system and can improve its adaptiveness and robustness (Tang et al. 2018).

For IM, shop-floor entities can be considered as agents based on the theory of multi-agent systems. These agents collaboratively implement dynamic reconfigurations to achieve agility and flexibility (Li, Tang, et al. 2017). This system requires **self-organised and collaborative architecture**, in which an individual machine should decide whether to change over production to another product type. Ma et al. (2020) propose anarchic manufacturing, a free market-based distributed planning and control system, delegates decision-making authority and autonomy to system elements at the lowest level to manage highly volatile and complex scenarios. Qian et al.

Table 2. Opportunities to develop IM.

Trend and event	CMarket opportunity	CTechnology opportunity
Digitalisation	• Investments in digital exchange relationships are increasing (Bokrantz et al. 2020a); • Customers are badly in need of smart products and services (Trstenjak et al. 2020); • Social media provide a knowledge-sharing platform for outsiders to comment on the IM implementation (Cordeiro, Ordóñez, and Ferro 2019); • Rapid experimentation and simulation of customers' ideas become possible (Cordeiro, Ordóñez, and Ferro 2019); • Companies can rapidly start businesses in market niches (Fuchs 2020; Mittal et al. 2020).	• Technology associated costs are reduced (Bokrantz et al. 2020a); • Digitalisation converses the simple analog data into computerised and integrated data (Qu et al. 2019; Seetharaman et al. 2019); • Manufacturers may overcome 'lacks' of modernity by using the new ICTs (Fuchs 2020); • The feedback from different departments easily realises bottom-up exchange of information (Cagliano et al. 2019; Cordeiro, Ordóñez, and Ferro 2019; Trstenjak et al. 2020); • The unit is more capable of designing and testing products rapidly by communications (Braccini and Margherita 2019; Cordeiro, Ordóñez, and Ferro 2019; Shi et al. 2020).
Servitisation	• Intelligent capabilities facilitate advanced services such as Smart Maintenance (Cenamor, Sjödin, and Parida 2017; Mueller, Buliga, and Voigt 2018; Cagliano et al. 2019; Büchi, Cugno, and Castagnoli 2020; Bokrantz et al. 2020a, 2020b); • Manufacturers are required to simultaneously enrich the value proposition by adding services while maintaining cost levels (Cenamor, Sjödin, and Parida 2017); • Servitisation inevitably connect the buyer and the seller and enable organisational changes such as more interdependent structures (Mueller, Buliga, and Voigt 2018; Seetharaman et al. 2019; Culot et al. 2020); • The servitisation process needs to be completely constructed based on IoT (Sanghavi, Parikh, and Raj 2019).	• A smart platform approach allows manufacturers to overcome the service paradox (Cenamor, Sjödin, and Parida 2017; Shi et al. 2020); • Servitised manufacturers are in great need of intelligent technologies that facilitate the design (Martínez-Olvera and Mora-Vargas 2019; Zhang, Ming, and Yin 2020), monitoring, and control of well-synchronized product-service-systems (Büchi, Cugno, and Castagnoli 2020; Wagire, Rathore, and Jain 2020; Liu, Feng, et al. 2020); • An effective interplay between digitalisation and servitisation effectively promotes financial performance (Kohtamäki et al. 2020).
Supply chain integration (globalisation)	• Under trading globalisation, customers prefer more complex and unique products in larger quantities, which resemble the demands in the IM system and these demands cannot be satisfied by a single provider (Benotsmane, Kovács, and Dudás 2019); • Manufactures with more teamwork, faster communication, closer connection, and better coordination promote internal and external integrations and accelerate the socialisation of products and processes (Bokrantz et al. 2020a); • Both supply chain integration and IM system need to be resilient to market competitions and rapidly changing demands (Benotsmane, Kovács, and Dudás 2019; Ramirez-Pena et al. 2020; Wang, Liu, et al. 2020).	• Technologies such as 5G, IoT, and blockchain are required to be applied in constructing integrated platforms and communication standards in complex collaboration networks (Lafferty 2019; Wagire, Rathore, and Jain 2020; Al-Sayed and Yang 2020); • Cloud manufacturing promotes supply chain integration by developing the superior quality of supplier relationships, collaborative partnerships, and improve customer satisfaction (Ge et al. 2020; Kamble et al. 2020; Wagire, Rathore, and Jain 2020); • Supply chain integration requires a great level of transparency of manufacturing processes and IT globalisation (Imran, ul Hameed, and ul Haque 2018; Shi et al. 2020; Wang, Liu, et al. 2020).
Green manufacturing	• Green Supply Chain Paradigm and popularity of Circular Economy connects the social aspects required at the IM systems (Braccini and Margherita 2019; García-Muiña et al. 2020; Ramirez-Pena et al. 2020); • Alongside IM enforcement, policies on green manufacturing and the renewal of resources are successively promulgated around the globe (Al-Sayed and Yang 2020; Won and Park 2020).	• Technologies used in IM systems such as Cloud Computing are also necessary to aid transformation to greener processes within organisations or along the supply chain (Jabbour et al. 2019; Ramirez-Pena et al. 2020); • In different stages of IM production, intelligence-based activities (such as preventive maintenance, accurate quality detection, and energy optimisation) save energy and reduce exhaust emissions from re-detection (Zhang, Ming, and Yin 2020).

(2020) propose an intelligent collaborative mechanism for global optimisation based on the negotiations on resource configuration among tasks. Wang et al. (2018) also design an intra-layered negotiation mechanism to

Table 2. Opportunities to develop IM.

Trend and event	CMarket opportunity	CTechnology opportunity
Emergent public events (such as trade war and Covid-19 pandemic)	• Manufacturers gradually realise the importance of ensuring cybersecurity, including protect systems, networks, and data from cyber-attacks and avoid too much external dependence on core technologies (Benotsmane, Kovács, and Dudás 2019); • Manufacturers widely pursue a high level of product and machine reconfigurations in the face of drastic demand changes (Fatorachian and Kazemi 2018; Büchi, Cugno, and Castagnoli 2020).	• The self-organising function required in the IM system (such as self-sensing and self-adaptivity) can solve the emergent requirements through the multi-agent and asynchronous backtracking technology (Qu et al. 2019).
Humanitarian relief	• During the purchase and production, customers and employees pay more attention to keeping personal health, safety, and human rights than ever before (Benotsmane, Kovács, and Dudás 2019; Chen 2019; Seetharaman et al. 2019; García-Muiña et al. 2020; Ramirez-Pena et al. 2020); • Personalised and intelligent products and services are popular in the markets (Jones et al. 2020; Saniuk, Grabowska, and Gajdzik 2020; Zhang, Ming, and Yin 2020).	• The development of human-machine interaction technologies and human-robot collaboration mechanisms in the IM system is in great need to guarantee employees safety at the workplace (Digiesi et al. 2009; Kamble et al. 2020; Wagire, Rathore, and Jain 2020); • Augmented Reality devices provide safety training for employees and increase employees' ergonomics (Mittal et al. 2020; Saniuk, Grabowska, and Gajdzik 2020); • Digital technologies easily provide customers with new digital options and highly personalised products (García-Muiña et al. 2020; Ge et al. 2020).

implement dynamic reconfiguration so that the system can support the hybrid production of multi-typed products by agents. Li, Tang, et al. (2017) propose intelligent evaluation and control algorithms to reduce the load-unbalance with the assistance of big data feedback in a multi-agent system.

From a self-organising perspective, Zhang et al. (2018) propose a framework depicting the mechanism and methodology of smart production-logistics systems to implement intelligent modelling of key manufacturing resources and investigate self-organising configuration mechanisms. In fixed-position assembly islands, Guo, Zhong, et al. (2020) develop a self-configuration model for autonomously configuring the optimal quantity of assembly islands and corresponding production activities based on the customer demand and production constraints. Liu, Zhang, et al. (2019) propose a self-organised architecture for resource configuration in platform-based IM systems to optimise the collaborative resource configuration across enterprises. Shi et al. (2019) realise the collaboration by modularity architectures. Wojtynek, Steil, and Wrede (2019) introduce an extended plug-and-produce concept based on a collaborative human-robot interaction scheme.

4.2.2. Modelling methods

The current dynamic production environment of IM leads to the configuration and reconfiguration of manufacturing systems. To achieve intelligent organising of resources, their requirements and the relationships between them need to be modelled. The literature proposes many models to address the optimisation of resource configuration, such as digital twins, multi-agent, ontology modelling, game theory, uncertainty theory, and mathematical programming. Table 4 categorises the relevant papers by different methods.

The literature shows that the multi-agent method is the most used to implement dynamic reconfigurations to achieve agility and flexibility. These agent models integrate advantages of central and distributed control and can achieve better load balancing performance (Li, Tang, et al. 2017). Many studies use the Ontology web language to model IM systems (Xu and Hua 2017; Deja and Siemiatkowski 2018; Wan et al. 2018; Järvenpää et al. 2019; Wan et al. 2019; Zhang, Xu, et al. 2021). Ontology can be used to map intelligent devices to the model instance and ensures that all intelligent participants involved in manufacturing understand each other (Wan et al. 2018). As seen from Table 4, studies on digital twins have been on the rise since 2019. Digital twins can achieve hardware-in-the-loop simulation of both physical equipment and the cyber model; this can be used to avoid the huge cost of reconfiguration if design deficiencies are found in the deployment process (Liu et al. 2020b). Digital twins can also realistically describe physical manufacturing resources, which allows reconfigurable strategies to improve the reusability of functions and algorithms and ensure the system satisfies the various requirements for different granularities and goals (Zhang, Xu, et al. 2021).

In addition to the above three methods, Qiu et al. (2005) investigate the possibility of using a non-cooperative

Table 3. Outcomes of IM implementation.

Possible outcome	Indicator	Typical reference
Economic performance	Profitability and competitiveness	Imran, ul Hameed, and ul Haque (2018); Braccini and Margherita (2019); Bokrantz et al. (2020a); Chu et al. (2020); Fuchs (2020); García-Muiña et al. (2020); Kohtamäki et al. (2020); Mittal et al. (2020); Yang, Ying, and Gao (2020)
	Productivity	Imran, ul Hameed, and ul Haque (2018); Moeuf et al. (2018); Benotsmane, Kovács, and Dudás (2019); Braccini and Margherita (2019); Buer et al. (2021); Ciano et al. (2021); Nimbalkar et al. (2020)
	Innovation performance	Coatney (2019); Felstead (2019); Kao, Nawata, and Huang (2019a); Wagire, Rathore, and Jain (2020); Yang, Ying, and Gao (2020)
	Flexibility	Fatorachian and Kazemi (2018); Moeuf et al. (2018); Braccini and Margherita (2019); Buer et al. (2021)
	Production cost	Moeuf et al. (2018); Al-wswasi, Ivanov, and Makatsoris (2018); Buer et al. (2021); Ge et al. (2020); Kamble et al. (2020); Kang et al. (2018); Romero-Silva and Hernandez-Lopez (2020)
	Product quality	Moeuf et al. (2018); Buer et al. (2021); Kamble et al. (2020); Kang et al. (2018)
	Production efficiency/ time	Moeuf et al. (2018); Imran, ul Hameed, and ul Haque (2018); Al-wswasi, Ivanov, and Makatsoris (2018); Buer et al. (2021); Ge et al. (2020); Kamble et al. (2020)
	Business stability and resilience	Braccini and Margherita (2019); Cordeiro, Ordóñez, and Ferro (2019); Kamble et al. (2020); Ramirez-Pena et al. (2020); Saniuk, Grabowska, and Gajdzik (2020)
	Customer satisfaction	Imran, ul Hameed, and ul Haque (2018); Cordeiro, Ordóñez, and Ferro (2019); Seetharaman et al. (2019); Ge et al. (2020); Shi et al. (2020)
Environmental performance	Energy consumption	Fatorachian and Kazemi (2018); Braccini and Margherita (2019); Koilo (2019); García-Muiña et al. (2020); Ramirez-Pena et al. (2020); Yang, Ying, and Gao (2020)
	Renewable resources employment	Benotsmane, Kovács, and Dudás (2019); Coatney (2019); Jabbour et al. (2019); Kao, Nawata, and Huang (2019a, 2019b); Koilo (2019); García-Muiña et al. (2020)
	Waste prevention	Braccini and Margherita (2019); Chen (2019); Buer et al. (2021); García-Muiña et al. 2020; Lukasik and Stachowiak (2020); Mittal et al. (2020)
	Water/air/land pollution prevention	Koilo (2019); García-Muiña et al. 2020
	Biodiversity	Braccini and Margherita (2019); Koilo (2019)
Social performance	Workplace safety	Braccini and Margherita (2019); Benotsmane, Kovács, and Dudás (2019); Ciano et al. (2021); Kamble et al. (2020); Mittal et al. (2020); Saniuk, Grabowska, and Gajdzik (2020)
	Workload of personnel	Braccini and Margherita (2019); Chu et al. (2020); Saniuk, Grabowska, and Gajdzik (2020)

game-theoretic technique for making reconfiguration decisions at the resource controller level in a highly competitive industry, where each player (i.e. resource controller) plays the game using the currently given options. Quantitative modelling is rarely used in this research area. Gen, Lin, and Zhang (2009) formulate a generalised mathematic model under the background of IM and apply evolutionary techniques to solve the configuration problem. Kumar, Baldea, and Edgar (2016) provide a bi-level programming framework for determining the optimal placement of sensors and actuators in the furnace.

4.2.3. Performance of resource configuration

Because it is difficult for information systems to collect complete information from the shop floor in IM systems, it is not appropriate for a centralised decision-maker to make a unified decision for resource configuration (Qiu et al. 2005). In a distributed decision system, it is important to identify the performance and goals for IM resource configuration. Many performance evaluation criteria have been proposed that can be divided into two categories: internal system performance and performance adapt to the external environment. The former includes system utilisation (Qiu et al. 2005; Schoenemann et al. 2015; Wan et al. 2018) and load balance (Li, Tang, et al. 2017; Rodrigues, Oliveira, and Leitão 2018), while the latter includes reconfigurability (Farid 2017), flexibility (Wan et al. 2019; Wojtynek, Steil, and Wrede 2019), and robustness (Wan et al. 2019; Tang et al. 2020).

System utilisation. IM systems make it possible to avoid starving and blocking to achieve high system utilisation while producing multiple product types with unknown demands and high volumes (Schoenemann et al. 2015). System utilisation is therefore an important criterion for evaluating the performance of resource (re)configuration. Under this criterion, an individual resource controller makes decisions to minimise the expected level of system backlog to achieve high system utilisation (Qiu et al. 2005). Ontology-based resource

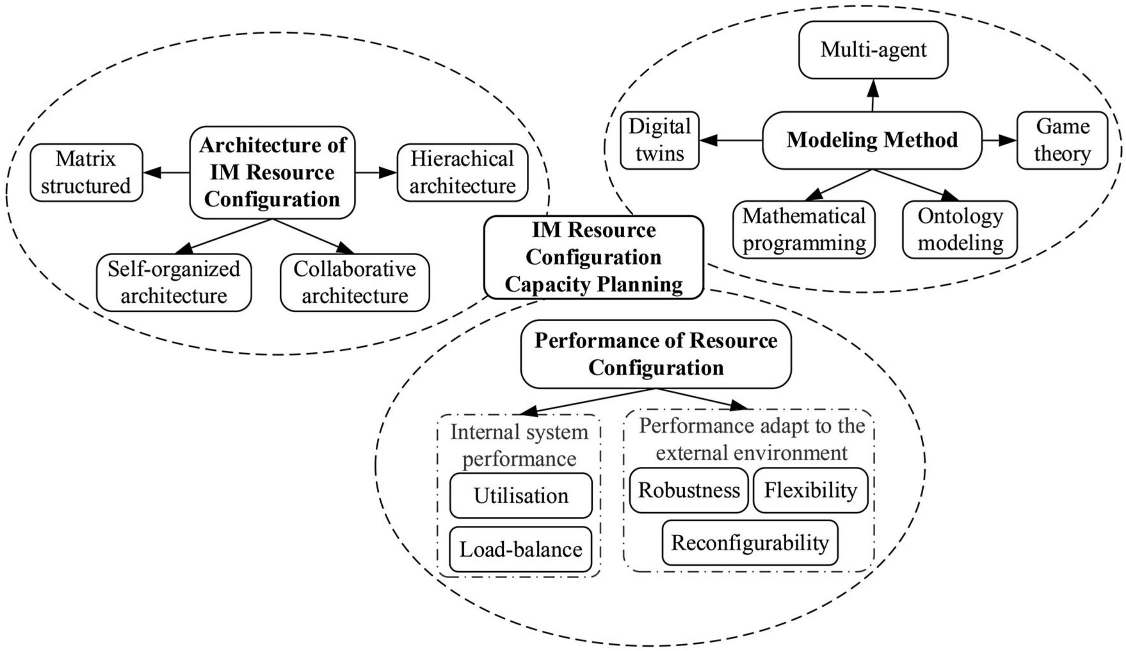


Figure 8. Framework for IM resource configuration and capacity planning.

Table 4. Modelling methods of IM resource configuration.

		Modelling methods					
Authors	Year	Digital twins	Multi-agent	Ontology modelling	Game theory	Uncertainty theory	Mathematical programming
Qiu et al. (2005)	2005				√		
Gen, Lin, and Zhang (2009)	2009						√
Oztemel and Tekez (2009)	2009		√				
Leitao, Barbosa, and Trentesaux (2012)	2012		√				√
Barenji, Barenji, and Hashemipour (2016)	2016		√				
Kumar, Baldea, and Edgar (2016)	2016						√
Li, Tang, et al. (2017)	2017		√				
Xu and Hua (2017)	2017		√	√			
Chien, Dou, and Fu (2018)	2018					√	
Wan et al. (2018)	2018			√			
Deja and Siemiatkowski (2018)	2018			√			
Rodrigues, Oliveira, and Leitão (2018)	2018		√				
Tang et al. (2018)	2018		√				
Järvenpää et al. (2019)	2019			√			
Leng et al. (2019)	2019	√					
Liu, Zhang, et al. (2019)	2019		√				
Wan et al. (2019)	2019			√			
Zhang, Xu, et al. (2021)	2019	√		√			
Liu, Jiang, and Jiang (2020)	2020	√					
Liu, Leng, et al. (2020)	2020	√					
Ma et al. (2020)	2020		√				
Tang et al. (2020)	2020		√				

reconfiguration facilitates reconfigurable development of manufacturing resources and can also greatly improve system utilisation.

Load balance. For IM systems with multiple entities and without global coordination, problems such as load imbalance and inefficiency may occur due to agent's different abilities and performance (Li, Tang, et al. 2017). It is therefore necessary to establish a criterion to evaluate load balance and efficiency after resource (re)configuration. Rodrigues, Oliveira, and Leitão (2018) show the benefits of applying the collaborative interaction protocol to balance the reconfiguration performed individually by the distributed agents and illustrate how well balanced is the production.

Reconfigurability. To meet the needs of highly customised products, IM systems require continuous resource reconfiguration. However, it is unclear to what extent these designs achieve the desired level of reconfigurability, which systems are indeed quantitatively more reconfigurable, and how these resources overcome their limitations to achieve greater reconfigurability in subsequent iterations. Farid (2017) develops a reconfigurability measurement process based on axiomatic design knowledge and a design structure matrix. These measures are then demonstrated to be valid for assessing reconfigurability requirements in automated and IM systems.

Flexibility. In the context of Industry 4.0, highly dynamic changing environments and uncertain personalised demands require quick responsiveness and flexibility of resource configuration in IM systems. Wan et al. (2019) aim at accomplishing functionality reconfiguration and flexible scheduling of manufacturing resources in pharmaceutical production. Robots operating in dynamic environments with constrained workspaces and exchangeable automation components also typically require flexible manufacturing processes. Wojtynek, Steil, and Wrede (2019) propose a unified modelling approach for resources in the digital-twin-based manufacturing system to improve efficiency and flexibility.

Robustness. In general, equipment failures are inevitable in manufacturing systems. Resource (re)configuration needs to be highly robust. Tang et al. (2020) show that a reconfigurable method based on their proposed architecture can be used in the mixed-flow production scenario based on random orders to improve robustness. Wan et al. (2019) propose a data-driven reconfigurable production mode to improve the system robustness.

4.3. IM production planning

Production planning is an important part of operations management and ensures smooth and orderly daily production activities. In a highly competitive market environment, manufacturers must undertake production planning to achieve a rapid response to dynamic and personalised demands. In the context of Industry 4.0, IM systems enable the interconnection of materials, machines, and people and encourage more flexible and agile production planning. There are many new intrinsic factors and external environments that motivate innovation in production planning for IM systems. This subsection reviews the related literature on motivation and intelligence features.

4.3.1. Motivations of intelligent production planning

International competition in the manufacturing sector is fierce, and most factories are networked and collaborative. Product customisation makes the production process complex, and demand fluctuates greatly, which aggravates the randomisation and dynamics of the manufacturing environment and increases the difficulty of making production plans. For traditional production planning, approximate or real-time dynamic decision-making is an arduous task, which promotes its intelligent (Schuh, Potente, and Hauptvogel 2014). Enterprises now not only take the interests as the goal but also consider energy conservation and environmental protection. Manufacturers should therefore consider the collaboration among multiple resources and multiple objectives when making production plans. This subsection analyses the motivations for IM production planning according to existing articles based on four aspects: random factors, dynamic environments, collaboration planning, and multiple objectives.

Random factors. Due to asymmetric information and abnormal interference in the manufacturing process, there are many random events, such as order quantity, raw material price, machine processing time, resource, and process randomness (Gasior and Józefczyk 2008; Hermann et al. 2019), which cause great difficulties for production management. Customers' demands are increasingly customised, and some scholars have considered this manufacturing mode (Zhang, Wang, et al. 2019; Olumide, Sgarbossa, and Strandhagen 2020). Manufacturers should not only carry out high-quality and low-cost production, but should also increase the flexibility of production planning to adapt to changes in customisation demands. However, customers also require low cost and fast delivery, leading to stronger demand for rapid response and self-optimisation in production planning (Sun, Huang, et al. 2020). Some studies use effective tolerance selection to reduce the need for rescheduling under disturbance events and increase planning robustness (Rossit, Tohmé, and Frutos 2019b).

Dynamic environments. Manufacturing plants are usually in an environment where internal and external conditions are dynamically changing, including machine states, production processes, customer demands, and supply chains (Stricker, Kopf, and Lanza 2014). In a dynamic environment, production replanning and resource reallocation are important issues (Hermann et al. 2019). Due to the limitations of real-time information access, traditional planning models usually make decisions under static environments. In IM systems, real-time data transmission and processing become possible, which provides technical support for autonomous decision-making. Rossit, Tohmé, and Frutos (2019b) set up the criterion to identify events that lead to the production replanning in advance to realise adaptability and self-optimisation of production decision-making.

Collaboration planning. Due to the increasingly competitive market and the fine division of labour, manufacturers are usually networked and collaborative in organising production. Multiple types of resources must also work collaboratively on different processes for the same product in the same factory. Nowadays, many intelligent technologies are applied to optimise collaborative resource allocation among enterprises by building a platform for aggregating distributed resources. It is thus increasingly necessary to make intelligent decisions for the collaborative production planning of multi-plants. Sun, Yamamoto, and Matsui (2020) study the horizontal integration of enterprise global value chain network and the production planning based on real-time production process data. Some studies jointly optimise production plans of multiple production lines or factories (Herrera, Thomas, and Vera 2013; Angizeh et al. 2020).

Multiple objectives. In recent years, manufacturers have considered not only economic objectives such as profit and cost but also sustainability, environmental protection, efficiency, and low energy consumption. Stricker, Kopf, and Lanza (2014) study multi-objective production planning in a dynamic environment considering the risk and uncertainty. Angizeh et al. (2020) propose a mixed-integer linear programming model for the production planning of multiple production lines by minimising the costs of production, energy, and labour. Moreover, governments around the world have paid increasing attention to environmental protection and energy conservation in manufacturing, and thus IM also has 'green manufacturing' requirements. Production planning decisions therefore usually require consideration of multiple interrelated and mutually restricted objectives.

4.3.2. IM production planning considering intelligent features

Combining with next generation ICTs. With the application of next generation ICTs in IM systems, manufacturers are committed to using new production methods with large amounts of real-time data. These new ICTs make it possible to access external demand data and internal production data, which provides an opportunity to evaluate and adjust complex production planning through data-driven methods. Some papers have tried to integrate production planning into the IM system using, for example, the IoT, machine learning, and cloud computing. Cloud manufacturing, a service-oriented manufacturing mode, requires networked production planning with information and resource sharing (Helo and Hao 2017; Hsu, Wang, and Chu 2018). Olumide, Sgarbossa, and Strandhagen (2020) propose a conceptual model with a product process framework for a smart production planning and control system combining these emerging technologies and capabilities.

Quick response to changes. In Industry 4.0, due to the dynamic changes in the market environment, IM systems must quickly adjust production planning according to real-time data (Zhang, Wang, et al. 2019). Timely response to dynamic events affects the delivery efficiency of the whole system. To ensure rapid-response production planning, some papers consider decision-making systems based on intelligent algorithms (Rodríguez, Gonzalez-Cava, and Pérez 2020). They consider the uncertainty of dynamic customer demand and internal interference and propose algorithms to support intelligent rapid decision-making.

Learning-based planning. Self-optimisation and self-adaptation based on learning are key features of IM systems. IM technology has created a more complex manufacturing system, involving all kinds of machines and decision-makers serving different processes. An intelligent production planning method needs to constantly enrich its knowledge based on real-time data in practice and through the use of this knowledge to make the next stage decisions. Rodríguez, Gonzalez-Cava, and Pérez (2020) propose a decision-making system based on fuzzy logic build on machine learning for production planning in the IM system.

4.4. IM scheduling

Scheduling is a critical decision problem in POM, which has a considerable influence on the efficiency and costs of IM systems. Due to the broad application of ICTs, as well as flexible and agile production organisation modes such as small-batch production and mass customisation production, manufacturing systems are currently being upgraded to distributed, dynamic, and intelligent systems. Zhang, Ding, et al. (2019) review the traditional job-shop scheduling studies and discuss their combinations with new ICTs under Industry 4.0. However, given the huge amount of data and great uncertainty and complexity in the IM context, traditional responsive scheduling can no longer adapt to the requirements of the required 'predictive' and 'agility' features of the IM system. The existing literature on IM scheduling adopts different methodologies to solve the scheduling problem under different settings. Thus, this section reviews the related literature from the perspective of problem formulation, methodologies, and real applications.

4.4.1. IM scheduling problem formulation

Traditional manufacturing scheduling mainly concerns the assignment of jobs to machines to minimise the makespan, energy, costs, and other objectives under certain constraints. IM scheduling controls management for new manufacturing systems such as CPS, cloud manufacturing, and digital twin systems and develops models for different production organisation modes with different objectives. This subsection reviews the related papers from the perspective of the manufacturing system and models for different production organisation modes with different objectives.

Manufacturing systems. Traditional production scheduling is concentrated on centralised scheduling or semi-distributed manufacturing systems. Under Industry 4.0, IM scheduling should deal with a smart or distributed system supported by novel and emerging technologies such as CPS, cloud manufacturing, and digital twins.

(a) *CPS.* Rossit, Tohmé, and Frutos (2019a) propose a data-driven architecture with a data engine using big data techniques to extract vital information generated from the CPS system. It allows predetermined scheduling arrangements through the utilisation of real-time information. Lin et al. (2018) propose an IoT-enabled real-time synchronisation approach for an automobile standard part factory with objectives of both horizontal synchronisations for the production consistency and vertical synchronisations for the coordination between adjacent processing stages. Jiang et al. (2018) put forward a two-layer distributed optimal rescheduling method for CPS system based on multi-agent systems with a dynamic decision cycle and a multistage negotiation mechanism through contract net protocol. Wu and Shen (2020) unify the definition of the planning, scheduling, and dispatching in the production system based on the CPS.

(b) *Cloud manufacturing.* Cloud manufacturing is a networked manufacturing mode that promotes the agile, service-oriented, green and intelligent development of the manufacturing sector. It faces challenges such as long scheduling time and latency or load imbalance among edge nodes due to the huge amount of data transmission and computation. Existing job scheduling algorithms can obtain local optimal solutions with a high computational cost, especially for large problem instances. Most papers concerning cloud manufacturing focus on solving these problems. Jian, Ping, and Zhang (2021) formulate a cloud edge-based two-level hybrid scheduling learning model and propose algorithms for fast prediction of the cloud-edge collaborative scheduling results. In a cloud environment, Yao et al. (2019) construct a performance estimation model and propose an intelligent scheduling algorithm by incorporating the simulated annealing into the genetic algorithm (GA). Huang and Huang (2013) develop a cloud computing-based IM scheduling system with the quality prediction method.

(c) *Digital twins.* Liu, Feng, et al. (2020) propose an intelligent scheduling method for workshops to formulate a scheduling scheme rapidly and efficiently by establishing a super-network model of a feature-process-machine tool in the digital twin workshop. Guo, Li, et al. (2020) propose a digital twin-enabled graduation IM system with real-time task allocation and execution mechanisms to achieve real-time information sharing and production planning, scheduling, execution, and control with reduced complexity and uncertainty.

Models for different production organisation modes with different objectives. To improve the flexibility and agility of manufacturing systems, IM usually applies production organisation modes such as hybrid flow shop (HFS), fuzzy job shop (FUJS), or flexible job shop (FLJS). Some papers combine scheduling decisions with maintenance decisions. Multi-objective scheduling problems are studied in many papers.

(a) *HFS.* Solano-charris, Montoya-torres, and Paternina-arboleda (2011) formulate an HFS scheduling problem with two stages for minimising the makespan and total job completion time and propose an ant colony optimisation procedure to solve the problem. Wang, Zhong, et al. (2016) propose a Petri net-based scheduling model for IoT-enabled HFS manufacturing, which switches the schedule of token sequences in the Petri net to the trigger sequence for transition. Gonzalez-Neira and Montoya-Torres (2017) propose a greedy randomised adaptive search procedure for the HFS scheduling problem. Fan, Yang, and Zhang (2019) propose a mutant firefly algorithm for a two-stage HFS scheduling problem. Framinan, Fernandez-Viagas, and Perez-Gonzalez (2019) investigate the utilisation of real-time information on job completion times in a flow shop with variable processing times for the rescheduling of multi-stage flow shop manufacturing systems.

(b) *FUJS.* In recent years, FUJS scheduling models have been widely studied. Jana et al. (2013) propose a dynamic task allocation mechanism for machine scheduling in a job shop environment by developing a scheduling method with a ratio analysis technique under the fuzzy multi-criteria decision-making environment. Sun et al. (2019) propose a hybrid cooperative coevolution algorithm for the minimisation of fuzzy makespan. Li, Liao, et al. (2020) formulate four comprehensive FUJS scheduling models with different criteria uncertainty and propose three scheduling algorithms to solve these models. Wen et al. (2020) study an integrated process planning and scheduling problem with uncertain processing time and due date; they formulate the integrated problem with fuzzy due date based on fuzzy set and propose a modified honey bees mating optimisation algorithm to solve the model.

(c) *FLJS.* Flexible scheduling is a significant technique for IM systems. Realisation of an optimised schedule through flexible resources assignment is critical to the application and popularisation of flexible scheduling, especially in uncertain manufacturing environments. Zhang et al. (2017) propose a dynamic and data-driven optimisation model for FJS scheduling based on game theory with real-time data, in which each machine is an active entity requesting tasks and processing tasks are assigned to optimal machines according to the real-time status. Ortiz et al. (2018) develop a dispatching algorithm for the scheduling and sequencing of FLJS systems in the textile sector. Zheng and Wang (2019) apply the epsilon-constraint method to solve a bio-objective FLJS scheduling problem to minimise the makespan and processing cost. Sang, Tan, and Liu (2020) establish a multi-objective FLJS intelligent scheduling model with complex constraints and propose an improved GA to solve the model.

(d) *Scheduling combined with maintenance.* Preventive maintenance is widely applied in industry to enhance production efficiency and machine reliability. Liao, Chen, and Yang (2017) formulate a joint problem considering the trade-off between maintenance cost and delivery date and propose an improved NSGA-II for parallel machines to find a compromise solution. Chen, An, et al. (2020) propose an accurate maintenance model based on reliability intervals combined with an FLJS scheduling problem and develop an improved NSGA to solve the model. Hu, Jiang, and Liao (2020) propose a joint optimisation method of job scheduling and preventive maintenance planning when the machine failure rate is time-varying under a fixed operating condition. Mi et al. (2020) propose a GA to solve the scheduling problem for predictive maintenance.

(e) *Multiple objectives.* In IM systems, manufacturers are under various requirements due to rapidly and dynamically changing market demands, which therefore require multiple objectives. Cost minimisation is critical for manufacturers to increase profits and improve competitiveness, including the operations costs, labour costs, delivery data-related costs, etc. (Bansal and Darbari 2012; Zhang, Song, and Wu 2013; Gong, Liu, et al. 2019). Flexibility and quick responsiveness are key features for IM systems, making timely completion and delivery another important objective (Zhang, Song, and Wu 2013; Manupati, Chang, and Tiwari 2016; Fan, Yang, and Zhang 2019; Tang et al. 2019; Gong, Liu, et al. 2019; Sang, Tan, and Liu 2020; Li, Yang, et al. 2020). Apart from the above-mentioned objectives, with the popularity of sustainable manufacturing, energy efficiency has been taking into great consideration (Gong, Liu, et al. 2019; Sang, Tan, and Liu 2020). Moreover, some papers also consider other objectives such as quality maximisation (Sang, Tan, and Liu 2020) and workload balance (Manupati, Chang, and Tiwari 2016; Sang, Tan, and Liu 2020).

4.4.2. Methodology for IM scheduling

In complex manufacturing environments, IM scheduling problems are usually non-deterministic polynomial-time hard with great complexity, and it is difficult to obtain promising solutions with traditional exact solution methods. Many papers propose meta-heuristic algorithms, learning-based methods, and agent-based methods.

Meta-heuristic algorithms are widely used for job shop scheduling problems; they are a type of bio-inspired algorithm that imitates the methods of nature, including the GA, ant colony algorithm (ACA), simulated annealing algorithm (SAA), etc. GA is one of the most widely used meta-heuristic methods and has many derivatives for different conditions (Pach et al. 2014; Oleśków-Szlapka and Pawlowski 2016; Manupati, Chang, and Tiwari 2016; Chen et al. 2020b; Liu, Li et al. 2020). ACA can be applied to different optimisation problems in complex manufacturing environments with time-varying and unpredictable characteristics (Solano-charris, Montoya-torres, and Paternina-arboleda 2011; Wang, Zhong, et al. 2016; Wu, Tian, and Zhang 2019). SAA is often used as a hybrid with other bio-inspired meta-heuristic methods (Rabiee et al. 2014; Wei et al. 2018; Tang et al. 2019).

Learning-based methods. *Reinforcement learning* is widely used for decisions in dynamic IM environments with real-time information. Bouazza, Sallez, and Beldjilali (2017) use a reinforcement Q-learning approach for the complex scheduling problem in smart factories. Liu, Li, Tang, et al. (2019) integrate big data analytics, reinforced learning, and optimal route planning methods for the operations in smart factories. Zhao et al. (2019) establish a reinforcement learning-based granulation and prediction process to determine the future trends of the gas tank level that targets real-time determination of the scheduling moment. Shiue, Lee, and Su (2020) propose a reinforcement learning-based approach for dynamic scheduling in a product-mix flexibility environment by incorporating two main mechanisms including an off-line learning module and a Q-learning-based module. Zhou, Zhang, and Horn (2020) propose a deep reinforcement learning-based method for dynamic scheduling to minimise total makespan in the IM system. *Deep learning* methods have been proved highly efficient and precise for forecasting in image, text, and other kinds of signals. Han and Yang (2020) combine a deep convolutional neural network with reinforcement learning to form a deep reinforcement learning framework for an order-oriented multi-stage sequential decision-making problem. Some studies use deep learning methods for dynamic machine parameters prediction to support real-time scheduling decisions (Essien and Giannetti 2020; Jian, Ping, and Zhang 2021).

Agent-based methods. Agent-based scheduling technology has long been considered important for industrial systems. Specifically, multiagent-based scheduling methods can dynamically and flexibly schedule manufacturing processes through cooperation and coordination among the agents in a distributed way. They are therefore widely used in IM systems (Walker, Brennan, and Norrie 2005; Guo and Zhang 2009a, 2009b, 2010; Sheng, Tang, and Lv 2010; Manupati, Putnik, et al. 2016). Multi-agent methods can also be used in combination with other techniques. To reduce the complexity of managing manufacturing systems and human interference and to improve resource utilisation and service quality, Madureira and Pereira (2011) combine autonomic computing, multi-agent systems, case-based reasoning, and bio-inspired optimisation techniques for supporting dynamic and distributed scheduling in manufacturing systems with autonomic computing properties. Malik and Kim (2020) integrate agent cooperation mechanism and fair emergency first scheduling scheme for optimisation of tasks execution, machines' resource utilisation, productivity, and delays with the consideration of exception.

4.4.3. IM scheduling implementation in real applications

Production planning and scheduling is not only a long-standing research area of great practical value, but also an acute industrial demand for manufacturers. Regrettably, most research results are seldom applied in industry, because existing methods can barely meet the requirements for practical applications due to the dynamic, complex, and distributed features of IM scheduling tasks in the real world. Some papers implement scheduling methods in real applications such as shop floor and mass customisation systems in various industries and some system prototypes (Shen, Lang, and Wang 2005; Aydin and Tunali 2009; Kang and Choi 2010; Yuan et al. 2014; Cupek et al. 2016; Ortiz et al. 2018). For the flexibility and generalisation of practicing, Ramadan et al. (2020) integrate lean tools and IM technologies to implement a dynamic value stream mapping lean framework and develop a smart lean-based real-time scheduling and dispatching module to achieve the lean target and minimise the manufacturing lead time. Apart from scheduling tasks, Cavalieri and Terzi (2006) implement a performance measurement system for the evaluation of scheduling solutions in an Italian brake manufacturer.

4.5. IM logistics

As a supporting system, IM logistics has not received as much attention as the other four themes. The related literature addresses logistics issues within an IM system (e.g. a smart factory) or among multiple IM systems (e.g. networked collaborative smart factories). We refer to the former as intra-factory logistics and the latter as inter-factory logistics. The existing literature adopts different methodologies to solve the issues in different scenarios, so we review the related literature from the perspectives of scopes and methodologies.

4.5.1. Scopes

IM logistics is not limited to a certain factory, but can include the whole supply network. Different scopes take different views on logistics issues.

Intra-factory logistics occur within the factory and especially on the shop floor. Most relevant studies focus on operational issues such as task scheduling, vehicle assignment, and routing. Cheng (2012) designs the core structure of the job shop scheduling system based on different rules, which may result in different travel distances of material flows. The rules include the mission-critical priority rule, first come first serve rule, the highest task priority rule, and the urgent tasks priority rule. Confessore, Fabiano, and Liotta (2013) focus on the problem of how to dispatch automated guided vehicles (AGVs) to execute different tasks when considering battery recharging and congestion or error issues. With the consideration of time window and carbon emission, Liao and Wang (2019) construct a collaborative production scheduling and logistics delivery model for a job shop under the background of green and IM. Bányai et al. (2019) study a matrix production system with flexible and configurable production and assembly cells which are arranged in a grid layout, where the material supply is based on autonomous vehicles. Xu et al. (2019) investigate logistics task scheduling in an intelligent factory workshop. Based on an application case of a coating chemical enterprise, Jiang et al. (2020) propose a digital-twin-based implementation framework for a production service system, which needs to respond to the personalised stochastic demand rapidly. Other scholars study design issues of intra-factory logistics. Mes, van der Heijden, and van Hillegersberg (2008) consider an agent-based logistics system, which adopts AGV to execute transportation jobs, for dough making process at an industrial bakery. They discuss how alternative system designs can be developed and compared. Gong, Zou, and Kan (2019) simulate an automobile mixed-flow assembly line to find its bottleneck. Funke and Becker (2020) predict flows of materials in factories. Hasan and Al-Rizzo (2020) investigate task scheduling for resource allocation in a job shop.

Inter-factory logistics typically involves supply chains that contain more general decisions, such as production and logistics planning and supply network design, compared to deciding task execution sequences in intra-factory logistics. To design a multi-echelon supply chain network, Bachlaus et al. (2008) integrate logistics decisions with production decisions from the perspective of strategic decision-making, such as assigning which cross deck to which distribution. Noticing the importance of efficient supply chain cooperation, Böhle, Dangelmaier, and Hellingrath (2009) integrate production planning with inbound logistics tour planning. Aissani et al. (2012) propose a model to adaptively schedule production and logistics tasks among companies located in different sites. De Falco, Mastrandrea, and Rarità (2019) focus on the architectures and dynamics of a scattered manufacturing network, where autonomous entities can share manufacturing resources, knowledge, and information, in the context of 3D printing. Lin, Qu, et al. (2020) propose a cloud-based production and logistics synchronisation service infrastructure for customised production systems. Oh and Jeong (2019) focus on a smart manufacturing supply chain to investigate its attributes and identify its functional and structural characteristics. Wang et al. (2019) capture routes with considerations of effects of route selection and complex geographical location of factories. Chen, Fang, and Tang (2020) focus on a cloud manufacturing platform that needs to allocate the endlessly emerging tasks to the resources scattered in different places. Pu et al. (2020) study a supply chain network reconfiguration in response to the rapid change of the market and the environment.

Combined intra- and inter-factory logistics. For the literature considering both logistics modes, scholars usually propose general frameworks. Ackermann and Muller (2007) explore the suitable models, planning procedures, and design solutions for a logistics network based on customer-oriented, directly linked, smallest autonomous service units. Jha, Babiceanu, and Seker (2020) propose a formal model for operational scheduling of cyber-physical resources for both intra-factory and inter-factory scenarios.

4.5.2. Methodologies

Typical methodologies for IM logistics include centralised mathematical programming, decentralised agent-based approach, and simulation.

Mathematical programming is a useful tool to help systems reach optimal performance. It is the most widely used methodology in IM logistics. Because IM logistics systems are complex, exact solutions are difficult to obtain. Heuristic and meta-heuristic algorithms are therefore proposed to solve the mathematical models. Bachlaus et al. (2008) use a multi-objective optimisation model to minimise the costs (fixed and variable) and maximises plant and volume flexibility. They propose a meta-heuristic algorithm based on particle swarm optimisation to solve the problem. To minimise the costs for producing, storing, and transporting the requested products, Böhle, Dangelmaier, and Hellingrath (2009) slightly modify capacitated lot-sizing problem and vehicle routing problem and propose a heuristic that can solve the problem in an acceptable time. Cheng (2012) designs different job shop schedule rules, such as the mission-critical priority rule, first come first serve rule, the highest task priority rule, and the urgent tasks priority rule. Follow these rules, a heuristic algorithm is developed. Confessore, Fabiano, and Liotta (2013) propose and exploit an ad-hoc heuristic algorithm for the application of the solution. To collaborate production and delivery, Hasan and Al-Rizzo (2020) propose a two-layer optimisation algorithm based on GA, which decomposes the model into two associated layers decision: the vehicle delivery and the workshop production scheduling decision. This problem decomposition idea is also used by Lin, Qu, et al. (2020). Xu et al. (2019) develop a mathematical model to minimise the finish time and the number of AGVs for an intelligent factory. They design a hybrid algorithm that integrates GA with ant colony algorithm. Compared to several benchmarks, the hybrid algorithm is effective on dynamic simultaneous scheduling. Oh and Jeong (2019) develop a profit-effective and response-efficient tactical supply planning model that balances the profit and lead time. Wang et al. (2019) build a model that defines the time, cost, and reliability attributes of logistics in more detail. They improve the artificial bee colony algorithm by introducing dimensional self-adaptation and group leader mechanisms to solve the model. Results show that the algorithm has advantages in searching capability, convergence speed, and solution quality. In the IM context, demand is stochastic and the system is complex, which makes many conventional search algorithms are not applicable. Chen, Fang, and Tang (2020) adopt the artificial neural network to predict the task completion status when allocating resources among candidates in real-time cloud manufacturing. Experimental results indicate that the artificial neural network-based scheduling approach outperforms the preferred method in terms of the optimisation objectives (e.g. total cost and makespan).

Compared to centralised mathematical programming, the decentralised **agent-based approach** is much more scalable and efficient as scale and complexity increase. To coordinate production and logistics among different factories, Aissani et al. (2012) propose a reinforcement learning-based multi-agent system where multiple agents could react with each other and effectively respond to disturbances. Different from Aissani et al. (2012), Mes, van der Heijden, and van Hillegersberg (2008) focus on the AGV system in a shop floor. They define five types of jobs that should be executed by AGVs according to the demands and design the agent system following 4 steps: decompose the functionalities based on the main goal, allocation the functionalities to agents, construct interactions between agents, and design details (e.g. job creation and evaluation, design waiting strategy, arrival and job schedule rules, and determine the price of jobs). To overcome the defects of the traditional supply chain model and to improve the flexibility, coordination, and integration of the supply chain network, Pu et al. (2020) focus on the supply chain resource reconstruction methods, environmental adaptation, negotiation, and coordination mechanisms.

Besides the agent-based approach, **simulation** is another useful tool that can handle complexity. Ackermann and Muller (2007) use a three-layer model to establish the network structure and use a combination of scenario-generation and optimisation to design the material flows in the network. They finally use simulation to analyse their designs. According to the characteristics of the mixed flow assembly, Gong, Zou, and Kan (2019) built a simulation platform of automobile mixed-flow assembly line to find out its bottleneck. For resource reliability and availability status, and logistics timing purposes, Jha, Babiceanu, and Seker (2020) use discrete state-machine diagrams to formalise the cyber–physical resource scheduling problem and build a simulation model to evaluate the solution performance and insights based on resource repair or replacement options, part transfer decisions. Jiang et al. (2020) propose a digital-twin-based implementation framework for the production service system. The system needs to respond to the personalised stochastic demand rapidly. The framework achieves autonomous decision-making control by synchronising production and logistics operations.

5. Discussions and future research directions

5.1. Discussions

As we mentioned in the introduction, to realise the desired benefits of IM, adapted POM approaches with key characteristics – integration, flexibility and networking, autonomous and collaborative decision-making, learning-based operations management, self-optimisation and adaptability, and proactive decision-making – need to be applied. This subsection analyses the 208 selected papers in terms of whether they have these characteristics to identify the trends of the current state of research. Table 5 presents the percentages of papers considering different characteristics in each research theme to the total number of papers in that theme.

5.1.1. IM value creation mechanisms

In publications related to value creation mechanisms, six IM characteristics are all mentioned as value creation factors or contexts. A few articles even regard these characteristics as outcomes of the IM system (Fatorachian and Kazemi 2018; Culot et al. 2020; Kamble et al. 2020; Mittal et al. 2020). Among the six, *integration* is the most frequently-cited characteristic. Scholars widely underline the importance of integrating physical (real world) and digital (virtual world) technologies such as intelligent machines, big data analytics, AI, IoT, 5G, and advanced robotics and engineering software to realise interoperability during the production process. Some treat the enterprise-wide integration of technologies as internal (horizontal) integration and emphasise the simultaneous expansion of external (vertical) integration such as collaboration with the external environment and parties to enable sharing and consolidation of heterogeneous sources of data, information, and knowledge along the value chain.

The second frequently mentioned characteristic is *flexibility and networking*, which indicates that the dynamic interactivity, openness to profound discussions, and verification of information between machines or workshops in face of changeable demand can reduce variation around a target value and improve interconnectivity between products and processes (Al-wswasi, Ivanov, and Makatsoris 2018; Al-Sayed and Yang 2020; Buer et al. 2021; Nimbalkar et al. 2020; Romero-Silva and Hernandez-Lopez 2020). Similarly, *autonomous and*

Table 5. IM characteristics and their applications.

Percentages of papers that consider the following characteristics to the total number in each theme	Value creation mechanism	Resource configuration and capacity planning	Production planning	Scheduling	Logistics
Integration	83.33%	6.45%	68.75%	55.56%	7.69%
Flexibility and Networking	63.33%	6.45%	56.25%	50.79%	15.38%
Autonomous and Collaborative Decision-Making	58.33%	25.81%	50.00%	31.75%	10.26%
Learning-Based Operations Management	21.67%	0.00%	6.25%	26.98%	2.56%
Self-Optimisation and Adaptability	53.33%	29.03%	31.25%	14.29%	12.82%
Proactive Decision-Making	6.67%	3.23%	12.50%	12.70%	5.13%

collaborative decision-making is equally important since they characterise the combination between the independent capability of physical entities (i.e. machines and other single components) ruling and controlling part of the IM system and taking autonomous actions in the face of the disturbed environment and their collaborative actions that tackle complex assignments in teams with others, in order to meet production objectives (Alwswasi, Ivanov, and Makatsoris 2018; Cagliano et al. 2019; Mittal et al. 2020; Ramirez-Pena et al. 2020; Liu, Feng, et al. 2020).

Self-optimisation and adaptability are often cited together with *autonomous and collaborative decision-making* since they provide the basis for autonomous and collaborative control (Martínez-Olvera and Mora-Vargas 2019). Some scholars also call the characteristic self-adaptability (Won and Park 2020). However, *Learning-based operations management* and *proactive decision-making* are seldomly mentioned in the publications as value creation mechanisms since their features are either overlapped with others or lack clear boundaries.

Although all six characteristics are covered in value creation researches, most publications merely describe the characteristics and their possible functions in the practice (Bokrantz et al. 2020b). Not much quantitative research assesses the magnitude and significance of how IM systems containing these characteristics affect the value creation path. Besides, the relationships and overlaps among these characteristics are required to be further clarified. Furthermore, how these characteristics are combined to affect a specific strategic planning, technology innovation, and operations and organisational management of the IM system lack deep investigation.

5.1.2. IM resource configuration and capacity planning

In publications related to IM resource configuration and capacity planning, all characteristics except *learning-based operations management* are mentioned. Among the five remaining characteristics, *autonomous and collaborative decision-making* and *self-optimisation and adaptability* are the most frequently cited and the other three are mentioned in only one or two papers. This is consistent with the fact that when multiple job streams compete for the same shared critical resource, it is extremely difficult for the resource controller to make effective reconfiguration decisions in a centralised manner (Qiu et al. 2005). Therefore, approaches with characteristics *autonomous and collaborative* and *self-optimisation and adaptability* are critical for IM resource configuration and capacity planning. Additionally, many papers use multi-agent architecture to model the configuration process to realise these two characteristics (Xu and Hua 2017; Li, Tang, et al. 2017; Deja and Siemiatkowski 2018; Wan et al. 2018; Järvenpää et al. 2019; Wan et al. 2019; Zhang, Xu, et al. 2021).

The literature that mentions *proactive decision-making* is associated with digital twins (Leng et al. 2019). Considering that the digital twin is an emerging concept in recent years, proactive decision-making may be a promising direction for IM resource configuration and capacity planning. A relatively small number of papers study *flexibility and networking* and *learning-based operations management*, which indicates that the current state of research lacks deep investigation to improve the flexibility and agility of IM systems to cope with highly dynamic environments.

5.1.3. IM production planning

In publications related to production planning, the six characteristics of IM are all mentioned. Among the six, *integration* is the most frequently cited. Most studies focus on the integration of various entities and the integration of various planning decisions. Many studies take system *integration* as an important research focus for IM and focus on vertical integration or horizontal integration of enterprises in IM systems. Vertical integration refers to the organic integration of the field, control, and workshop management layers, ensuring that information can be transmitted to production resource planning (Zhang, Wang, et al. 2019; Lee, Ju, and Woo 2020). Accordingly, production planning based on the vertical integration of enterprises needs to integrate the information of multiple links in the manufacturing environment to realise the whole process of intelligent decision-making. Horizontal integration refers to the integration of intelligent systems in different manufacturing stages, including the configuration of materials, energy and information within a company, as well as the configuration of value networks between different companies. Consequently, production planning based on horizontal integration needs to integrate information from each stage of the system.

The second frequently mentioned characteristic is *Flexibility and networking.* IM systems require flexibility for production planning in the network structure. On the one hand, the flexibility of easily adjusting the production plan according to market changes is necessary to adapt to the uncertainty in the production process. On the other hand, many enterprises are adopting a multi-factory cooperative production mode to speed up order delivery, making it necessary to develop a collaborative production plan. Due to the variety of customer demands and regional environments, networked production planning is becoming a hot research topic in IM (Sun, Yamamoto, and Matsui 2020). Thus, the joint effects of the production and market environments promote flexible and networked planning.

Autonomous and collaborative decision-making and *self-optimisation and adaptability* are also frequently mentioned. The autonomous decision-making capabilities help IM system to realise real-time planning and control. With the strengthening interactions in IM systems, scholars establish collaborative decision-making in real-time communication and interaction (Helo and Hao 2017). In addition, combined with adaptive decision-making, production planning can realise self-optimisation in a complex and changeable production environment. Some papers consider unexpected events in the production environment to improve the adaptability of production planning (Stricker, Kopf, and Lanza 2014; Rossit, Tohmé, and Frutos 2019b) or use fuzzy numbers to represent randomness in production to realise the self-optimisation and self-adaptation of the plan through the parameter adaptive strategy (Rodríguez, Gonzalez-Cava, and Pérez 2020).

The characteristics *learning-based operations management* and *proactive decision-making* are mentioned in only one and two papers, respectively. Machine learning methods integrate expert knowledge to realise machine decision-making with inaccurate and uncertain control data (Rodríguez, Gonzalez-Cava, and Pérez 2020), and support IM to realise intelligent decisions and optimisation based on data. The core feature of IM is to enrich the knowledge base, complete autonomous learning, and optimise decision-making.

5.1.4. IM scheduling

In publications related to scheduling, all six characteristics of IM are mentioned. Among the six, *integration* is the most frequently cited. Most scheduling frameworks are based on CPS through the integration of physical (real world) and digital (virtual world) technologies (Lin et al. 2018; Rossit, Tohmé, and Frutos 2019a) or cloud manufacturing systems with the integration of regional manufacturing resources and services (Yao et al. 2019; Jian, Ping, and Zhang 2021). Some papers emphasise the importance of integration between planning and scheduling (Wu and Shen 2020), while others integrate different algorithms for better performance (Rabiee et al. 2014; Wei et al. 2018; Tang et al. 2019; Han and Yang 2020).

The characteristic *flexibility and networking* is also frequently mentioned, which indicates efficient use of manufacturing resources. Most of the IM scheduling models are flexible, with multi-functional machines and multi-stage scheduling (Solano-charris, Montoya-torres, and Paternina-arboleda 2011; Wang, Zhong, et al. 2016; Sun et al. 2019). Manufacturing resources can also be re-allocated and inter-connected in a service-oriented manner through cloud computing systems to match various manufacturing demands flexibly and efficiently.

Self-optimisation and *adaptability* are frequently mentioned along with *collaborative decision*-making and *Proactive decision-making* in the distributed architecture. In this area, the multi-agent approach is widely used to solve scheduling optimisation problems under complex conditions through coordination and negotiation mechanisms among agents, thus enabling proactive, collaborative, and adaptable decisions (Guo and Zhang 2009a, 2009b; Sheng, Tang, and Lv 2010; Jana et al. 2013; Manupati, Chang, and Tiwari 2016; Manupati, Putnik et al. 2016; Zhao et al. 2019; Malik and Kim 2020). Agents represent the basic processing entity to meet the demand for flexibility in building models within the complicated IM system.

Learning-based operations management is intensively studied in the publications focus on learning-based algorithms because they are believed to represent the future trends for IM scheduling. Reinforcement learning is the most widely used learning-based algorithm for dynamic scheduling problems (Shiue, Lee, and Su 2020; Zhou, Zhang, and Horn 2020). In addition to reinforcement learning, many other learning-based algorithms such as convolutional neural networks and long short-term memory are applied in this area (Essien and Giannetti 2020; Han and Yang 2020; Jian, Ping, and Zhang 2021).

Although all six characteristics are covered in scheduling research, how these characteristics can be combined to form an IM scheduling method requires further consideration. How to apply these approaches to management systems to solve practical problems also requires deeper investigation.

5.1.5. IM logistics

In publications related to IM logistics, all characteristics are mentioned. From Table 5, we can see that *flexibility and networking* is the most intensively studied characteristic and mainly focuses on the design of the supply chain structure and how to increase the supply chain's flexibility in response to rapidly changing demands (Ackermann and Muller 2007; Bachlaus et al. 2008; Confessore, Fabiano, and Liotta 2013; Oh and Jeong 2019; Wang et al. 2019; Gong, Zou, and Kan 2019). Because of the new generation ICTs, all entities are integrated within the IM system. There is still potential to increase the scale of integration by incorporating more processes to achieve overall optimality.

The IM system is complex and large. It involves handling material within multiple subsystems and fluctuating, online customised demands. Traditional methods based on mathematical programming cannot be used to address the logistics problem within or among smart factories. Therefore, *self-optimisation and adaptability* and *autonomous and collaborative decision-making* are also frequently studied (Mes, van der Heijden, and van Hillegersberg 2008; Aissani et al. 2012; Cheng 2012; Bányai et al. 2019; De Falco, Mastrandrea, and Rarità 2019; Xu et al. 2019; Gong, Zou, and Kan 2019; Jha, Babiceanu, and Seker 2020; Jiang et al. 2020; Pu et al. 2020).

Integration is important for IM because the factories constitute a network, and different systems within a factory are interconnected. Unlike traditional research that only considers one procedure of manufacturing, studies on IM integrate multiple separate procedures and optimise the system according to different objectives (Böhle, Dangelmaier, and Hellingrath 2009; Liao and Wang 2019; Lin, Qu, et al. 2020). Some studies also consider *flexibility* and *proactivity* in IM. Based on big data analysis, it is possible to forecast demand and realise *proactive decision-making*, which can improve the responsiveness of the system (Funke and Becker 2020; Hasan and Al-Rizzo 2020). To improve prediction accuracy, some learning-based operation management methods such as neural networks have been proposed (Chen, Fang, and Tang 2020).

5.2. Future directions

5.2.1. Future directions for value creation mechanisms

IM management involves a collective intelligence of technology, organisation, human resources, knowledge, products, and services (Martínez-Olvera and Mora-Vargas 2019). Existing studies on IM value creation are mostly concept-oriented. They concentrate on the clear definitions of IM stages, features, and outcomes. Although this study has extended the landscape to the delineation of the PPP for IM implementation, future value creation research needs to be detailed in the following three aspects through the use of more quantitative methodology or delicate case studies.

Strategic planning of IM adoption and implementation. Although governments around the world have enacted initiatives, action norms, and policies to promote intelligent transformations and earlier practitioners have obtained some first-mover advantages (Büchi, Cugno, and Castagnoli 2020), IM adoption and implementation still require careful blueprinting and strategic planning because it is a long-term transformation involving several components (Fuchs 2020). The strategic vision can be extended to address the following questions:

- How should benchmark measurement for different stages of IM implementation be set?
- Does social and environmental performance substitute or complement the economic performance of IM implementationïij§Are there any avoidable trade-offs in the Triple Bottom Line during the IM transformation?
- What are the differences between IM implementation in different contexts such as different industries, scales, firm sizes, and previous experiences?
- Given the institutional and cultural differences, how do we make comparisons between IM adoption and implementation in developing and developed countries?
- How does IM interact with other strategies to affect performances (e.g. low cost, differentiation, and digitalisation)?
- How can IM transformation make full use of the developing opportunities described in Section 4.1.2 (such as servitisation, green manufacturing, supply chain integration, etc.) to obtain a long-term performance?

Technology innovation and information system management in IM systems. Technology transformation has been widely regarded as the main aspect of IM transformation (Cordeiro, Ordóñez, and Ferro 2019), and the lack of research has clarified the technology evolving process of these new ICTs and how they affect innovation diffusion in IM systems. There are thus significant opportunities to investigate the following questions:

- How can we make accurate technology predictions on the new generation ICTs that are widely used in IM such as AI, 5G, or IoT?
- How do we balance the ambidextrous innovations in the technology adoption (i.e. exploration and exploitation of the new ICTs) of IM systems? How do the two types of innovations affect the performance of IM systems?
- How do the technology innovations in an IM system diffuse to another? What are the innovation diffusion mechanisms inside and outside the manufacturing company?
- Does IM transformation accompany radical/incremental innovation on the business model? How do the business models interact with technologies to affect the performance of IM implementation?

Organisational transformation and operations management of IM systems. IM implementation cannot be realised only with leading technologies. Appropriate organisational and operations management is required (Kohtamäki et al. 2020). Some necessities, abilities, and opportunities have been listed in section 4.1. These influencing factors are too fragmented to guide the real practice directly unless causality-based value creation paths are constructed. Thus, there are some basic research questions about organisational and operations management of IM systems to address:

- Do IM systems have to transform every business activity (e.g. product design, facility layout, logistics and distribution, marketing, and supplier selection) to be autonomous, intelligent, and smart? What are the performance differences between the firm with complete IM transformation and those with partial transformationsïij§
- Humans and machines have their own strengths in accomplishing different tasks, and their relationships are attracting more academic attention (Cordeiro, Ordóñez, and Ferro 2019; Jerman, Bach, and Aleksic 2020). How to evaluate the complementary effect and enhancing effect of human-machine collaboration in different environments and tasks in human-machine collaborative systems? What is the human-machine collaboration mechanism?
- How to express the human learning effect and its impact on system performance? Does the IM system reduce human participation or provide more jobs?
- Who is appropriate to be the governor of the IM transformation (government, incumbent CEO, or external experts)?
- Within IM innovation ecosystems, how do different roles (such as innovator, financial agents, researchers, etc.) collaborate? Are the organisational structure and relationship among these roles static or dynamic?

5.2.2. Future directions for IM resource configuration and capacity planning

Existing studies on IM resource configuration and capacity planning primarily focus on framework, model, and performance analysis, most of which are concept-oriented and qualitative. In practice, the dynamically changing and uncertain manufacturing environment puts forward new requirements for IM resource configuration that need self-adaptation and real-time learning in new conditions, as well as self-organisation to complete dynamic resource configuration and achieve a rapid response to change. Future research could be expanded in the following three aspects.

Production organisation mode and product portfolio decisions for the IM system. IM systems need to be highly flexible and adaptable. Facing the increasingly segmented demand of multi-variety and small-lot customisation, manufacturers need to optimise their decisions for production organisation and product portfolios to achieve rapid responses to market demand at low cost.

- Product demand patterns include high volume, multi-variety, small volume, and customisation. How should production organisation modes be organised such that they are compatible with the characteristics of the manufacturing products and their demand patterns to determine the appropriate production or procurement mode for finished products, assemblies, components, and raw materials?
- Given product and demand characteristics such as production structure and quantity, how can production organisation decisions be optimised with multiple objectives of production delivery and costs?
- How to optimise the batch production of product generalisation modules for customised demand and mixed batch production of multiple products?
- In a human-machine collaborative IM system, how to configurate the human and machine capacities to improve the efficiency and flexibility and reduce costs? How to consider the human learning effect of workers with different proficiency?

The method of self-organised dynamic resource configuration. Based on the interconnection of manufacturing resources in IM systems, research is needed to achieve agile and real-time (re)configuration methods for complex manufacturing systems to meet dynamic changing demands. This should include consideration of the following.

- The resource organisation mechanism from three perspectives: production unit construction, production unit reconfiguration, and resource parameter change.
- Methods of coordinating manufacturing resource conflicts caused by simultaneous reconfiguration of multiple production units.
- Decision-making methods for optimal matching of resources and demand to support the capacity planning and reconfiguration of production units.

Big data analytics and intelligent algorithms. With continuous advances in computing technology and increased device connectivity, shop floor information could be collected more completely in the near future (Qiu et al. 2005). There are significant opportunities to use these advantages to enable resource configuration and capacity planning decisions in terms of the following aspects.

- Mining and automation of decision-support knowledge from interaction contexts are major challenges for achieving self-organised mass individualisation in IM systems (Leng et al. 2019).
- Dynamic events such as machine failures, tool shortages, and requirement variations make the problem more challenging. Future work should be conducted to incorporate big data analytics and intelligent algorithms while solving real-time configuration optimisation problems.
- Performance analysis in current research is mainly based on simulation or prototype systems, which are limited in terms of anomaly handling and errors due to the limited collected data (Leng et al. 2020). Experiments that are closer to a real-world situation are needed to improve the accuracy (Qian et al. 2020). Virtual and augmented reality might help to model and react to unpredictable objects and complex tasks, simulating the movement of resources with superimposed information from the real world.

5.2.3. Future direction for IM production planning

The increasing randomness and complexity of the production environment pose a huge challenge to intelligent production planning in IM systems. At the same time, production planning is also challenged by emerging technologies and must evolve with the support of new generation ICTs to meet existing intelligent management needs. Currently, although some studies have investigated production planning by intelligent algorithms such as machine learning, from an operational perspective, production planning challenges in the smart manufacturing environment have not been fully addressed. Future research directions for production planning include the following areas.

Deep integration of production planning and new generation ICTs. New ICTs allow for more data availability and more efficient system execution and feedback. It is critical to leverage these advantages when making production planning decisions to improve the flexibility, adaptability, and agility of IM systems. The deep integration of production planning and these technologies is important for future research.

- Study the method for predicting product demand based on machine learning using accessible massive data and characterising its stochastic pattern.
- Investigate machine learning methods for predicting the changing patterns of production capacity of various manufacturing resources based on industrial big data (e.g. data on equipment operation, personnel work) from CPS, taking into account the impacts of learning effects, equipment ageing and failure, manufacturing resource load, and production line reconfiguration.
- Combined with supplier data, study the method of predicting the manufacturing lead time for each product and component under different product demand and factory capacity conditions based on reinforcement learning.
- Study the production planning methods combining the traditional operations research methods and new generation ICTs such AI, big data, and machine learning.

Production planning methods for new production organisation modes. With the globalisation of the market and increasing competition, more manufacturers are adopting networked supply chain systems to organise their production. In addition, for increasingly personalised product demand, how to organise customised production quickly and efficiently at low cost is also an important issue. Therefore, there are promising research directions to propose effective production planning methods for these new production organisation modes.

- In IM systems, customer orders may be produced collaboratively by multiple factories that are distributed in a network. Different factories have different capabilities and capacities, costs, and objectives. Materials and components may also be provided as distributed

by different suppliers in a networked distribution. In IM systems with dynamically changing environments and data accessibility, it remains an interesting problem to propose a networked collaborative production planning method considering the overall efficiency and the interests of different subjects.
- In the customised manufacturing mode, many products have an uncertain bill of material structure, manufacturing lead time, and other key parameters at the time of order receipt. These need to be determined progressively as the design progresses. To reduce the delivery time, intelligent production planning methods need to be studied to meet such production organisation modes.
- In the human-machine collaborative systems, humans are flexible, creative, and can work with complicate and unexpected situations, while machines possess high speed, heavy workload, accuracy, and tirelessness. Given different production orders, how to make the production plan and the working plan of human workers and machines?

Production planning methods for practical IM systems. Current studies on IM production planning are mostly theoretical and exploratory and lack practical manufacturing scenarios. Numerical experiments are largely not based on real manufacturing cases (Zhang, Wang, et al. 2019). For example, the assumptions of the determined processing time, machine failure, maintenance time, demand, and simple customer requirements and objectives make the proposed production planning methods are not applicable in practical manufacturing systems. Therefore, it is necessary to study practical production planning methods for real IM systems, including answering the following questions.

- How to improve the feasibility, flexibility, and robustness of the production plan for practical manufacturing IM systems with complex and uncertain parameters?
- How to realise self-optimisation and self-decision-making in a dynamically changing IM environment for intelligent production planning?
- How to achieve collaborative optimisation of the production planning with other operations decisions such as procurement, outsourcing, and logistics planning in practical complex IM scenarios?

5.2.4. Future directions for IM scheduling

The features of *connectivity* and *transparency* of CPS-based IM systems provide a good foundation for IM scheduling in terms of data, while their *predictive* and *agility* features place new requirements for studies on IM scheduling.

Data-driven, real-time adaptive scheduling and rescheduling methods. In the context of Industry 4.0, enabled by the CPS and digital twins, information including demand, resources, production exceptions, machine breakdown, and supply chains becomes available. One potential topic includes research on real-time adaptive (re)scheduling methods based on massively available data to achieve rapid responses to system input changes, random system perturbations, and rapid changes in the environment.

Predictive and proactive scheduling considering future uncertainty. Uncertain disturbance events are inevitable in the production process, including equipment failure, urgent order insertion, and quality problems. These factors can make it impossible to continue the original scheduling plan. To improve scheduling efficiency and performance, studies taking advantage of data acquisition in CPS-based IM systems to make predictive and proactive scheduling decisions should be considered for further research.

IM scheduling based on human-machine collaboration. For the human-machine collaborative decision-making, some practical IM scheduling problems are so complex that it is difficult to make good decisions by relying only on human or computer algorithms. Thus, human capabilities can be introduced into the computational loop of intelligent systems. Coupling human cognition and the ability to analyse fuzzy and uncertain problems with a computerised intelligence system with strong computational and storage capabilities will constitute an enhanced intelligence form of ‘$1+1>2$’. For the human-machine co-working, given multiple human works and machines with various skills, how to make the task scheduling decisions to improve the complementary and enhancing effects for the human-machine collaborative IM systems?

5.2.5. Future directions for IM logistics

Logistics is an indispensable part of the IM system and has been widely studied. To meet the requirements of dynamic resource organisation and highly agile production planning and scheduling, intelligent logistics must be transformed accordingly. Future studies on the following research directions are still required.

Open and agile IM logistics system architecture. An open and agile logistics system can realise rapid and low-cost reconfiguration of logistics resources for different production organisation modes and material and product transportation tasks. Therefore, it is an important research direction to study the architecture of logistics resource self-organisation and task description methods for IM logistics systems.

Logistics operations in coordination with production planning and scheduling. IM systems have a wide range of production tasks involving the transportation of materials and components within the system as well as product transportation between systems. These transportation tasks are linked to the processing tasks of the machines. Therefore, to coordinate product manufacturing and logistics operation plans, studies on autonomous and coordinated optimisation methods should be considered for future research.

Adaptive scheduling methods for IM logistics based on new generation ICTs. The increasing demands for customised products have brought many inevitable operational dynamics and disruptions to the production logistics system, such as changes in production quantity, production plans, manufacturing processes and disruptions such as logistics equipment failure, system deadlock, and path congestion. New generation ICTs such as digital twins may be useful to monitor such dynamics and disruptions. Based on a virtual twin in the digital world, we can proactively test many reactions in response to complex and time-varying environment. Future studies could explore adaptive scheduling methods and applications based on ICTs in IM logistics.

6. Conclusions

IM is the future trend for today's market environment and technological condition. The main economies around the world have promulgated development strategies related to IM. These have brought new opportunities and challenges to research in the area of IM POM. In the current stage, it is necessary to identify the current state of the literature and the potential future directions for IM POM. To our knowledge, there is no review paper that discusses POM studies in IM. This paper thus conducts a systematic literature review of POM studies in the area of IM.

First, we selected 208 research articles published between 2005 and 2020 in this area following a systematic search process. Descriptive analyses of these selected papers are performed across their year of publication, authors' countries, and industrial sectors. The results show that the number of papers in this area has been growing rapidly in recent years, especially after the promulgation of relevant policies in various countries. In terms of the authors' country of origin, 86.6% of papers have their origins in Europe and Asia. From the perspective of industrial sectors, most papers do not indicate a specific industry. Among other papers with specific industries, the electrical equipment, computer and electronic products, transportation equipment, and machinery manufacturing industries are the most widely studied.

We performed thematic analysis of the selected papers based on five research themes: value creation mechanisms, resource configuration and capacity planning, production planning, scheduling, and logistics. We conducted in-depth analyses of different themes from the perspectives of PPP, architecture, motivation, methodology, and application. The research trends were identified for the existing related literature.

Finally, we discussed the current status, limitations, and challenges of existing research from the perspective of six characteristics of IM POM. Based on the above analyses and discussions, future research directions for various research themes were proposed. We believe this review can provide important and interesting ideas and insights to researchers in relevant areas. Concerning the implications for practice, the findings and insights of this research could provide general support for improving the application of POM-related methods in practical manufacturing management.

Despite its contributions, there are some limitations in this paper. First, we might have omitted some potentially relevant studies due to terminological differences and our focus on journal papers in the six databases, although we followed a systematic search process as presented in Section 2. Second, we mainly conducted the review according to five research themes in POM studies, so studies related to other POM topics may have been neglected.

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Data availability statement

The data that support the findings of this study are available within the article; supplementary data are available from the corresponding author (Zhibin Jiang) upon reasonable request.

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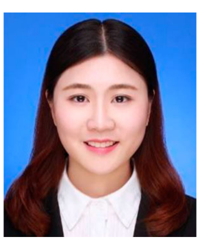

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Appendix

Table A1. List of papers in each theme.

Themes	Papers
Value creation mechanism	Digiesi et al. (2009); Cenamor, Sjödin, and Parida (2017); Al-wswasi, Ivanov, and Makatsoris (2018); Brizzi et al. (2018); Buer, Strandhagen, and Chan (2018); Fatorachian and Kazemi (2018); Imran, ul Hameed, and ul Haque (2018); Kang et al. (2018); Kasapoglu (2018); Kumar et al. (2018); Moeuf et al. (2018); Mueller, Buliga, and Voigt (2018); Benotsmane, Kovács, and Dudás (2019); Braccini and Margherita (2019); Cagliano et al. (2019); Chen (2019); Coatney (2019); Cordeiro, Ordóñez, and Ferro (2019); Felstead (2019); Haseeb et al. (2019); Jabbour et al. (2019); Kao, Nawata, and Huang (2019a, 2019b); Koilo (2019); Lafferty (2019); Martínez-Olvera and Mora-Vargas (2019); Qu et al. (2019); Sanghavi, Parikh, and Raj (2019); Seetharaman et al. (2019); Shi et al. (2019); Al-Sayed and Yang (2020); Bokrantz et al. (2020a, 2020b); Büchi, Cugno, and Castagnoli (2020); Buer et al. (2021); Chu et al. (2020); Ciano et al. (2021); Culot et al. (2020); Fuchs (2020); García-Muiña et al. (2020); Ge et al. (2020); Jerman, Bach, and Aleksic (2020); Jones et al. (2020); Kamble et al. (2020); Kohtamäki et al. (2020); Kovacova et al. (2020); Lin, Sheng, and Jeng Wang (2020); Liu, Feng, et al. (2020); Lukasik and Stachowiak (2020); Mittal et al. (2020); Nimbalkar et al. (2020); Ramirez-Pena et al. (2020); Romero-Silva and Hernandez-Lopez (2020); Saniuk, Grabowska, and Gajdzik (2020); Trstenjak et al. (2020); Wagire, Rathore, and Jain (2020); Wang, Liu, et al. (2020); Won and Park (2020); Yang, Ying, and Gao (2020); Zhang, Ming, and Yin (2020)
Resource configuration and capacity planning	Qiu et al. (2005); Gen, Lin, and Zhang (2009); Oztemel and Tekez (2009); Leitao, Barbosa, and Trentesaux (2012); Schoenemann et al. (2015); Barenji, Barenji, and Hashemipour (2016); Kumar, Baldea, and Edgar (2016); Farid (2017); Li, Tang, et al. (2017); Xu and Hua (2017); Chien, Dou, and Fu (2018); Deja and Siemiatkowski (2018); Rodrigues, Oliveira, and Leitão (2018); Wan et al. (2018); Tang et al. (2018); Wang et al. (2018); Zhang et al. (2018); Järvenpää et al. (2019); Leng et al. (2019); Liu, Zhang, et al. (2019); Wan et al. (2019); Wojtynek, Steil, and Wrede (2019); Shi et al. (2019); Zhang, Xu, et al. (2021); Guo, Zhong, et al. (2020); Leng et al. (2020); Liu, Jiang, and Jiang (2020); Liu, Leng, et al. (2020); Ma et al. (2020); Qian et al. (2020); Tang et al. (2020)
Production planning	Gasior and Józefczyk (2008); Böhle, Dangelmaier, and Hellingrath (2009); Herrera, Thomas, and Vera (2013); Stricker, Kopf, and Lanza (2014); Helo and Hao (2017); Hsu, Wang, and Chu (2018); Leusin et al. (2018); Hermann et al. (2019); Rossit, Tohmé, and Frutos (2019b); Angizeh et al. (2020); Lee, Ju, and Woo (2020); Olumide, Sgarbossa, and Strandhagen (2020); Rodríguez, Gonzalez-Cava, and Pérez (2020); Sun et al. (2020); Sun, Yamamoto, and Matsui (2020); Zhang, Wang, et al. (2019)
Scheduling	Shen, Lang, and Wang (2005); Walker, Brennan, and Norrie (2005); Cavalieri and Terzi (2006); Walker, Brennan, and Norrie (2006); Aydin and Tunali (2009); Guo and Zhang (2009a, 2009b); Guo and Zhang (2010); Kang and Choi (2010); Sheng, Tang, and Lv (2010); Madureira and Pereira (2011); Solano-charris, Montoya-torres, and Paternina-arboleda (2011); Bansal and Darbari (2012); Huang and Huang (2013); Jana et al. (2013); Zhang, Song, and Wu (2013); Hsieh and Lin (2014); Pach et al. (2014); Yuan et al. (2014); Rabiee et al. (2014); Wang et al. (2015); Cupek et al. (2016); Manupati, Chang, and Tiwari (2016, Manupati, Putnik, et al. 2016); Oleśków-Szlapka and Pawlowski (2016); Wang, Zhong, et al. (2016); Gonzalez-Neira and Montoya-Torres (2017); Bouazza, Sallez, and Beldjilali (2017); Liao, Chen, and Yang (2017); Zhang et al. (2017); Jiang et al. (2018); Lin et al. (2018); Ortiz et al. (2018); Wei et al. (2018); Fan, Yang, and Zhang (2019); Framinan, Fernandez-Viagas, and Perez-Gonzalez (2019); Gong, Liu, et al. (2019); Liu, Li, et al. (2019); Rossit, Tohmé, and Frutos (2019a); Sun et al. (2019); Tang et al. (2019); Wu, Tian, and Zhang (2019); Yao et al. (2019); Zhao et al. (2019); Zheng and Wang (2019); Chen, An, et al. (2020); Essien and Giannetti (2020); Guo, Li, et al. (2020); Han and Yang (2020); Hu, Jiang, and Liao (2020); Jiang et al. (2020); Li, Liao, et al. (2020a); Li, Yang, et al. (2020); Liu, Li, et al. (2020); Liu, Chen, et al. (2020); Malik and Kim (2020); Mi et al. (2020); Ramadan et al. (2020); Shiue, Lee, and Su (2020); Wen et al. (2020); Wu and Shen (2020); Sang, Tan, and Liu (2020); Zhou, Zhang, and Horn (2020)
Logistics	Ackermann and Muller (2007); Bachlaus et al. (2008); Mes, van der Heijden, and van Hillegersberg (2008); Böhle, Dangelmaier, and Hellingrath (2009); Aissani et al. (2012); Chan, Chung, and Chan (2012); Cheng (2012); Hussain et al. (2012); Lin et al. (2012); Louly, Dolgui, and Al-Ahmari (2012); Sabar, Montreuil, and Frayret (2012); Confessore, Fabiano, and Liotta (2013); Zhang and Anosike (2012); Delgado Sobrino et al. (2013); Wang et al. (2014); Drakaki and Tzionas (2016); Bortolini et al. (2017); Karimi and Bashiri (2018); Schmid, Limère, and Raa (2018); Bányai et al. (2019); De Falco, Mastrandrea, and Rarità (2019); Gong, Zou, and Kan (2019); Huh et al. (2019); Liao and Wang (2019); Oh and Jeong (2019); Ni, Wang, and Chen (2019); Wang et al. (2019); Xu et al. (2019); Chen, Fang, and Tang (2020); Funke and Becker (2020); Hasan and Al-Rizzo (2020); Jha, Babiceanu, and Seker (2020); Jiang et al. (2020); Kalempa et al. (2020); Li, Barenji Ali, et al. (2020); Lin, Qu et al. (2020); Lyu et al. (2020); Pu et al. (2020); Zhao, Lu, and Yi (2020)